

General Executive
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T H E M E S

Secondary Education

Third Year

Sections: General Sciences - Life Sciences

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Center for Educational Research and Development



Republic of Lebanon

Ministry of National Education, Youth and Sports

T H E M E S

Secondary Education

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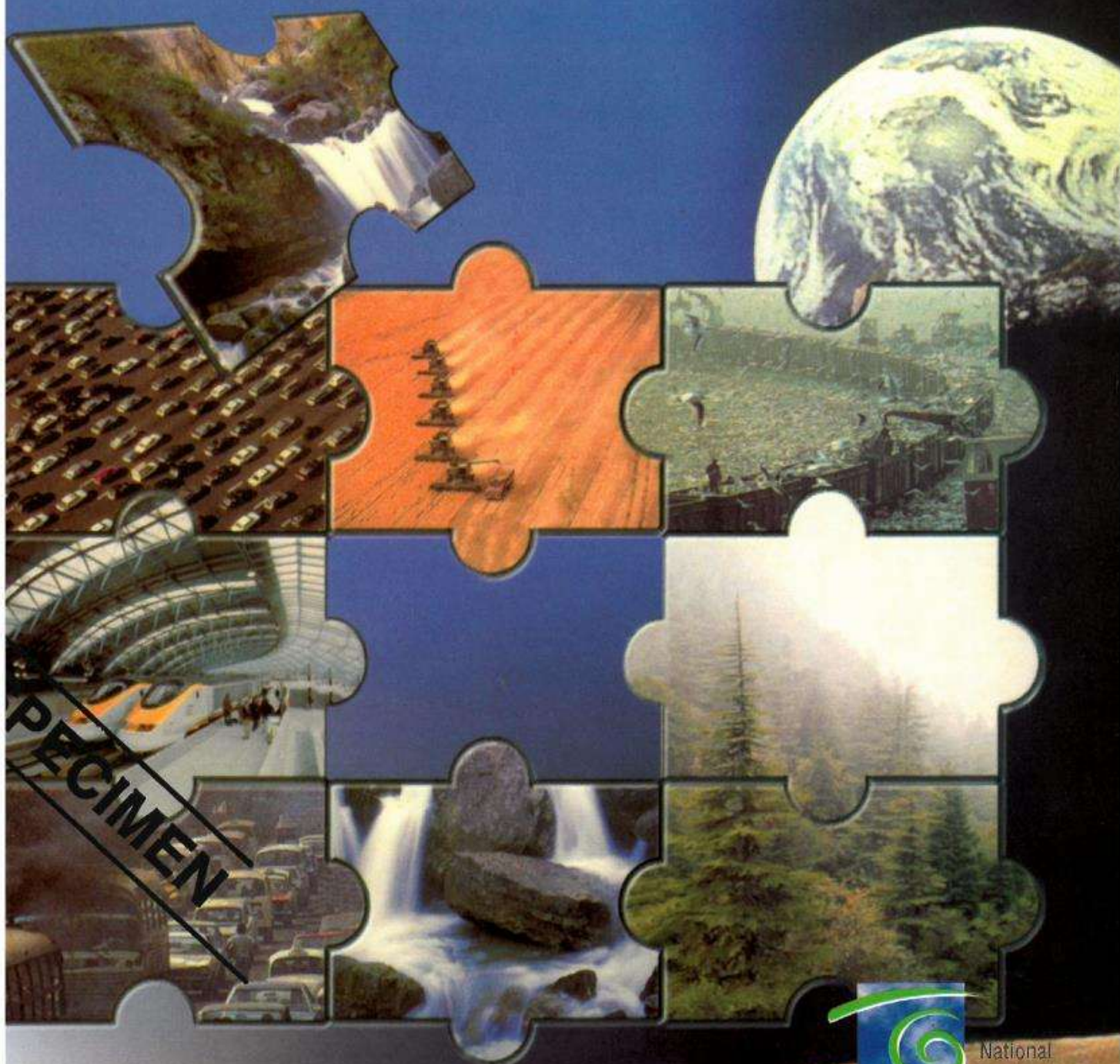
National
Textbook
New Curricula

THEMES

Secondary Education

Third Year

General Sciences - Life Sciences Sections



PECIMEN

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National
Textbook

New Curricula

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The National Textbook Project

By issuing the textbooks for the third year of each educational cycle, the Educational Center for Research and Development will have completed the third and last installment of books called for by the New Curricula. We are placing these books in the hands of students with the great hope that we are moving, step by step, toward the goal of acquiring sound learning, using sophisticated educational means and up-to-date methodology that encourage and reinforce individual thinking and research, acquisition of skills, development of ethical and national attitudes, the feeling of national belonging as well as the feeling of belonging to humanity at large.

The on-going revolution in information, communication and educational-aids technology has undoubtedly limited the role of the textbook and lowered the rank it used so recently to occupy. However, in our society and in many other societies, the textbook remains the basic means of education, and it is our duty to exert our utmost effort and care to come up with the best product as to form and content. Yet we should not lose sight of the fact that the textbook is not sufficient by itself but should rather be used as a stepping stone to access other sources of information. What is important is to keep a clear vision and maintain the right course toward our objective. The means should not turn into the end and the student should always remain the focus of the learning/teaching process.

No one ignores or denies the fact that textbook writing requires very high academic and educational qualifications and very wide field experience. The authors committees undeniably possess such qualities. Yet the textbooks of the last two years contained some negative aspects. Such is the nature of human work, no matter how good the intentions or how great the effort extended. Here constructive criticism constitutes a real contribution to raising the standard of authorship, minimizing errors and filling gaps. We say that, with all due appreciation and respect to all those who have contributed to the success of this project.

The Educational Center for Research and Development is embarking this year on a process of evaluating the New Curricula and related textbooks, teacher training courses and student achievement. This is a natural and necessary step now that the new system has been put into effect. This process aims at identifying the curricular objectives that have been achieved as well as those that have not been achieved, with a view to proceeding with the positive aspects and correcting the negative ones.

As part of this correction process, we plan to review the versions that have been issued in order to secure good textbooks for our students, who always deserve the best.

March 13, 2000

President, Educational Center for
Research and Development

Nemer FRAYHA

The time will come when diligent research over long periods will bring to light things which now lie hidden. A single life time, even though entirely devoted to the sky, would not be enough for the investigation of so vast a subject. And so this knowledge will be unfolded only through long successive ages. There will come a time when our descendants will be amazed that we did not know things that are so plain to them...many discoveries are reserved for ages still to come, when memory of us will have been effaced. Our universe is a sorry little affair unless it has in it something for every age to investigate. Nature does not reveal her mysteries once and for all.

-- Seneca (AD 100)

To the Student

Themes is a series of textbooks designed for students studying English as a first foreign language. This is the third book for the Secondary Cycle in the New National Curriculum.

Themes follows a thematically organized, content-based approach, which emphasizes process-product writing. Three thematic units are grouped into two major parts. Part I: The Natural World focuses on natural phenomena (E.S.P., tectonic plates, the Continental Drift theory, and Mars). Part II: The World of Human Intervention deals with technology (the Mind Candy Kafe, electronic robbery, telemedicine, biotechnology, and the geosciences) and current concerns (health, the environment and over-population).

The Themes series employs an integrated approach, whereby the student's listening, speaking, reading, writing, cultural awareness, critical thinking and study skills are developed. In this textbook, all the skills are consolidated in process-product writing. The textbook includes authentic selections. It is hoped that this approach will assist students in developing their language and furthering their communicative proficiency.

The series emphasizes learner-centered instruction, in which students are encouraged to express their opinions orally and in writing. Activities involve learners in individual, pair and group work.

Out-of-class assignments include reports, projects and research work. Sub-skills such as grammar and vocabulary have been integrated as tools used to develop the major skills. Vocabulary items needed for a better understanding of the selection are given in glossaries preceding the reading texts.

The authors believe that learning a language is a cognitively-based skill involving application of the language in tasks that relate to the student's academic and socio-cultural environment. It is through such an approach that one can acquire the necessary proficiency and competence to perform successfully. Learning a language can be an exciting and rewarding experience.

The Authors

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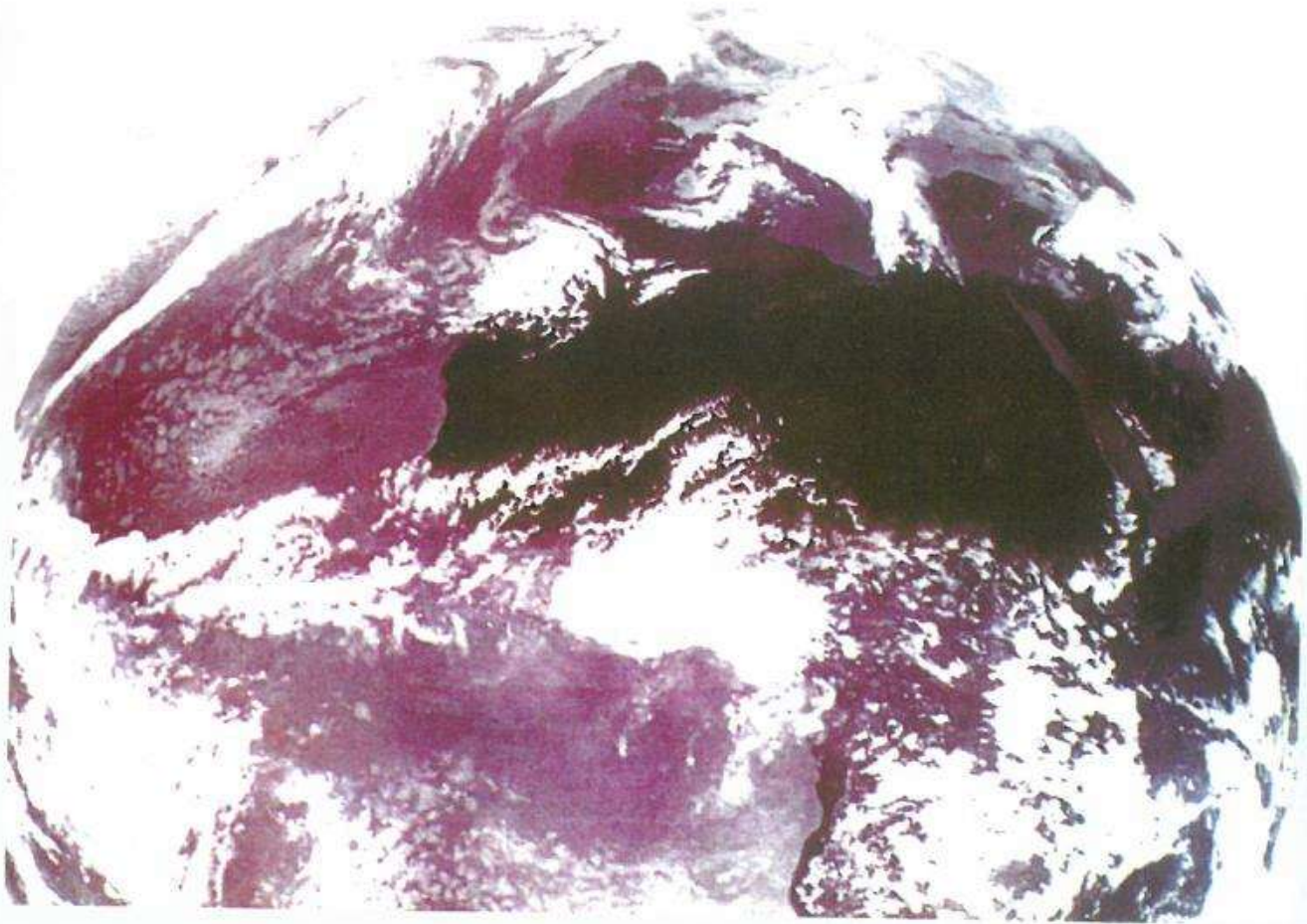
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THE NATURAL WORLD



Unit I: Natural Phenomena

"Rest is not idleness, and to lie sometimes on the grass under the trees on a summer's day, listening to the murmur of water, or watching the clouds float across the sky, is by no means a waste of time."

Sir J. Lubbock

"God reveals Himself in the orderly harmony of what exists."

Albert Einstein

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bolstered: v. supported.

cant. but a statistical analysis revealed that a difference that large among so many participants would seldom occur by chance.

Bolstered by such experiments, believers in ESP accuse research psychologists of the same sort of skepticism that led eighteenth-century scientists to laugh at the idea that meteorites come from outer space. They complain that today's skeptical scientists resemble the Italian philosophers who refused to look at Jupiter's moons through Galileo's telescope because they "knew" that such moons did not exist. Stubborn skepticism sometimes blinds people to surprising truth.

ORAL DISCUSSION

1. What are "astrological predictions"? Give your own examples of the three varieties of ESP: telepathy, clairvoyance, and precognition.
2. Do you agree with any of the three varieties? Have you witnessed any yourself?
3. What, in your opinion, is wrong with ESP?

ORAL DISCUSSION

1. Do you feel that there are "indeed any people who can read minds, see through walls, or foretell the future"?
2. How would you describe your attitude towards ESP: skeptical, believing, or other? Discuss with the rest of the class.
3. Explain: "We are creatures whose minds are tied to our physical brains and whose perceptual experiences of the world are built of sensations."

The preposition *at* is sometimes used with nouns without an article, such as *at random*. What is the difference between *at school* and *at the school*. What would you say if someone is in the office using *at*: My Dad is ...

READING

Claims of ESP

Claims of paranormal phenomena include astrological predictions, psychic **healing**, **reincarnation**, communication with the dead, and out-of-body flyer programs. Of these, the most respectable and testable claims are for three varieties of ESP:

Telepathy, or mind-to-mind communication—one person sending thoughts to another or perceiving another's thoughts.

Clairvoyance, or perceiving remote events, such as sensing that a friend's house is on fire.

Precognition, or perceiving future events, such as a political leader's death or a sporting event's outcome.

Closely linked with these are claims of psychokinesis, or "mind over matter," such as **levitating** a table or influencing the roll of a die.

On stage, the psychic, like a magician, controls what the audience sees and hears. In the laboratory, the experimenter controls what the psychic sees and hears. Consider one careful experiment where the experimenters had a computer generate a 100-item list **at random** of the digits 1, 2, 3, 4, and 5 for each of their 179 student participants. They gave each student such a list in a sealed envelope and asked the student to guess which number was in each of the 100 positions. By chance, 1 guess in 5, or 20 guesses out of the 100, should be correct. When told beforehand that ESP was **beneficial**, subjects averaged 20.66 correct out of 100. When told that ESP was harmful, they averaged only 19.49 correct. The difference might seem **insignifi-**

READING STRATEGIES: Glossary

healing: v. curing;

making someone healthy.

reincarnation: n. living again with a body after death, according to certain religious beliefs.

levitating: v. causing something to rise and float in the air as if by magic.

at random: without any set plan.

beneficial: adj. helpful; useful.

insignificant: adj. not of value or of importance.

Chapter 1

NATURAL PHENOMENA

The World Within Us - Part A

Perception and Sensation

WARM UP

1. How do we sense the world around us?
2. Other than providing sensory information, what does the brain supply us with?
3. What does ESP stand for?

How can you tell the meaning of words such as *wonder*, *roll* and *die* as opposed to *wander*, *role* and the meaning of *die* as singular of *dice* instead of the opposite of *live*?

READING STRATEGIES: Glossary

is saturated: v. is very full of.

discerning: v. noticing or understanding with difficulty.

die: n. singular of dice
skeptical; British.
spelling: sceptical;
doubting; distrusting.

READING

Is There Perception without Sensation?

Many people say they believe in extrasensory perception; others are not sure. The media **is saturated** with reports of psychic wonders: crimes solved, dreams come true, futures foretold. Paranormal television programs and series (such as "Unsolved Mysteries" and "The X-Files") and movies (such as *Close Encounters*) are big business. Are there indeed any people who can read minds, see through walls, or foretell the future?

In laboratory experiments, parapsychologists - those who study paranormal (literally, beyond the normal) happenings - have sometimes been astonished at psychics who seem capable of **discerning** the contents of sealed envelopes, influencing the roll of a **die**, or drawing a picture of what someone else is viewing at an unknown remote location. But other research psychologists and scientists are **skeptical**. If ESP is real, we would need to overturn the scientific understanding that we are creatures whose minds are tied to our physical brains and whose perceptual experiences of the world are built of sensations. Sometimes new evidence does overturn our scientific preconceptions.

ESP

WARM UP

Which do you consider more scientific : foretelling the future using a crystal ball or foretelling the weather?



Note the word *awestruck*. It is constructed from the phrase *struck with awe*. What other words do you know that are structured in the same way?

READING STRATEGIES: Glossary

gullible: adj. easily tricked.
hoax: n. a trick.
defy: v. to challenge.
awestruck: adj. filled with amazement.

READING

Skepticism about ESP

ESP skeptics claim that an uncritical mind is a **gullible** mind. They point out, time and again, that the so-called psychics have exploited unquestioning audiences with amazing performances in which they appeared to communicate with spirits, read minds, or levitate objects only to find out that the psychics' acts were a **hoax**, nothing more than the illusions of stage magicians. Indeed, many psychic deceptions have been exposed by magicians, who resent the exploitation of their arts in the name of psychic powers.

Even scientists get deceived. A notable case involved two teenage magicians, Steve Shaw and Michael Edwards. In 1979, this young pair visited Washington University's new parapsychology laboratory, offering to demonstrate their "psychic powers." For three years, the two pretended to **defy** the laws of nature. They appeared to project mental images onto film, cause clocks to slide across a table, effortlessly bend metal objects, and move objects in sealed jars. Forewarned against trickery by the youngsters' magician-advisor, the laboratory director nevertheless ignored the warnings and for a time proclaimed that the kids were reliable.

This parapsychologist's gullibility illustrates how tempting it is to label phenomena we do not understand as unexplainable. The **awestruck** observer would ask what explanation could there possibly be other than ESP. Thus, before bats' echolocation ability was discovered, many people attributed bats' ability to avoid wires in

complete darkness to clairvoyance. When the bats were blinded, when their noses were sealed, when their wings were coated with varnish, they could still navigate, so what other explanation could there be but ESP?

ORAL DISCUSSION

1. Explain the expression "time and again" in paragraph 1.
2. Have you ever watched anyone do any of the tricks done by the two teenagers Shaw and Edwards? Tell the class.
3. What is meant by "bats' echolocation ability" in the last paragraph?
4. What kind of question is the last sentence in the selection? Does the author expect an answer?
5. Can you provide your own reasons for being skeptic about psychic powers?

The prefix *dis-*, which generally carries a negative meaning, such as in *dismayed*, gives the reader an indication of the meaning of the whole word even if the stem is not clear. What other words do you know which start with *dis-*, carry a negative meaning, and have an unclear stem?

READING

Thinking Critically about ESP

In the past, there have been all kinds of crazy ideas—that bumps on the head reveal character traits, for example. When faced with such claims—or with claims of mind-reading or out-of-body travel—how can we separate crazy ideas from those that sound crazy but are true? At the heart of science, we find a simple answer. Test them to see if they work.

This scientific attitude agrees with believers and skeptics: what parapsychology needs to give it credibility is a reproducible phenomenon and a theory to explain it. Knowing how easily people are deceived, and lacking reproducible results, most research psychologists remain skeptical. Indeed, they are **dismayed** by all the shows, books, and magazines on paranormal topics. But some are often **intrigued** by reports of activities or situations that cannot be explained by today's scientific knowledge.

The scientific attitude blends curious skepticism with open-minded **humility**. It demands that extraordinary claims be supported by clear and reliable evidence. As critical thinkers, this can be our attitude, too. We can be skeptical but without closing ourselves to all startling claims. We can be open to new ideas without being gullible, discerning without being **cynical**. We can be critical thinkers; yet, knowing that our understanding of nature is incomplete, we can agree with Shakespeare's Hamlet that "there are more things in heaven and earth, Horatio, than are dreamt of in your philosophy." Some things that we assume to be true (the existence of God,

READING STRATEGIES: Glossary

dismayed: v. feeling upset.

intrigued: v. greatly interested

humility: n. the quality of being humble

cynical: adj. seeing little or no good in anything

life after death, etc.) are beyond science. That is why we retain a humble sense of wonder regarding life's untestable mysteries.

reverence: n. great respect and admiration

To gain deep **reverence** for life and to be in awe of life, we need look no further than our own perceptual system and its capacity for organizing formless nerve impulses into colorful sights, vivid sounds, and evocative smells. Within our ordinary perceptual experiences lies much that is truly extraordinary—surely much more than has so far been dreamt of in our psychology. A century of research has revealed many of the secrets of sensation and perception, but for future generations of researchers, there remain profound and genuine mysteries to explore.

ORAL DISCUSSION

1. Can you cite other examples of "crazy ideas" referred to in the first paragraph?
2. Explain what "a reproducible phenomenon and a theory to explain it" means.
How can you apply it, say, to Newton's gravity theory?
3. In no more than 40 words, summarize the last paragraph.

Language Conventions: Fragments

What are fragments? Fragments are incomplete sentences. Unlike speech, we should try to avoid them when writing formally. The following are typical examples of fragments. In each case, these are just bits and pieces of complete sentences.

1. Clause – Because of one's fascination with mysterious phenomena.
2. Phrase without a subject or verb – Being skeptical and almost impossible to convince.
3. No main verb – E.S.P. and its deep mysteries.
4. Transition expressions – For example, a skeptic who does not believe in extrasensory perception.

Directions: With a partner, find and correct any fragments in the following passage.

Why do people believe in something that can be scientifically proven wrong? Because they may want a simple explanation for a coincidence. For example, a woman planting a tree in her yard and the weather turning warm for the rest of the month. She reasons that planting trees causes warm weather. A simple, obvious conclusion. The weatherman, on the other hand, gives a long, confusing explanation. Such as "various meteorological factors caused displacement of the cold front." As a result, the woman who will believe her own explanation. Being simple and easily understood. Once one person believes this conclusion, others will believe it too.

Like the advertiser above, a writer must please his/her audience. If you identify your audience early in your writing, you will be better able to make assumptions about what they already know, do not know, or want to know. This will enable you to make better decisions about

whether to write formally or informally,
how careful to be about correct spelling, punctuation and grammar,
how to organize your ideas, and
what kind(s) of explanation(s) to make.

Thus, knowing something about your reader's background and expectations will help you become a better writer.

B. Focusing on the Purpose

Why are you writing? Are you writing to (1) entertain, (2) demonstrate your knowledge about a subject, (3) persuade someone to agree with your point of view, (4) present a new way of looking at a subject, or (5) inform your audience about a subject that they do not know much about but are probably interested in? Your purpose in writing is related to the type of writing you are doing. Try to identify the possible purpose(s) for the following types of writing.

Type of Writing

Possible Purpose

- a. a short story
- b. an article entitled "Thinking Critically About ESP"
- c. an essay praising a new but unpopular invention
- d. an article on how to survive the aftermath of an earthquake

Deciding on your purpose before you write will help you decide on *what* to write and *how* to write it.

Focusing on Process/Product Writing

Section 1: Writing Topics



"Write like a reader and read like a writer." What is the connection between reading and writing? Will becoming a better writer also make you a better reader or vice versa? In parts D and E, you will go through a process which will help you learn to write better in a more relaxed frame of mind.

First of all, as a writer, you must take into consideration who will be reading your writing – your audience. In most cases, it will probably be your teacher and/or your peers, but your readership may also include others in the school community or the society in which you live. In this case, you must take into consideration the background of your reader in order for your writing to be better understood and appreciated.

A. Focusing on the Audience

Who are you writing for? Pretend that you are an advertiser trying to persuade the following type of people to eat at your favorite restaurant. What features or facilities would you focus on?

Audience

- a small child
- a teenager
- a couple with a large family
- an older couple

Focus

WRITING SKILLS: PARAPHRASING

For your writing assignment in the chapter, you will be asked to write a paragraph on a topic related to natural phenomena. You may want to report someone else's material or ideas in your writing. Although direct quotations can be used in longer pieces of writing like essays, they do not usually appear in pieces of paragraph length. Therefore, a good writer often uses paraphrasing, which is about the same length as the original with changes in the wording and the sentence structure.

The following method may be used to paraphrase another author's ideas.

1. Select a sentence from a passage that you would like to quote in your work. Read the whole context of the sentence (the lines before and following) several times until you are sure that you understand the meaning.
2. Cover up the sentence and try writing down the substance of the sentence in your own words.
3. Check the original. If you have left something important out, add it.
4. If you used exactly the same words as the original, change them. Do not change the original meaning.
5. State the source so that you do not plagiarize.

Directions: Practice paraphrasing the following sentences using the method described above. The first sentence has been done for you.

"...we are creatures whose minds are tied to our physical brains and whose perceptual experiences of the world are built of sensations."

In the text "Is There Perception Without Sensation," the author states that there is no perception other than the physical senses which we have.

1. "Stubborn skepticism sometimes blinds people to surprising truth."
2. "ESP skeptics claim that an uncritical mind is a gullible mind."

Now compare your paraphrases with a partner.

In order to state sources, you can use one of the following suggestions.

In _____, _____ informs us that _____.
claims that _____
indicates that _____
shows us that _____
states that _____

3. Identifying a specific area: the Internet

Although these topics are narrowed, they are still not specific enough to write a research paper on. You need to further narrow your topic or make it more specific to cover an aspect of the topic you are dealing with.

4. More specific: surfing the Internet.

Sequenced Writing Project

Step 1: Choose a topic.



Step 2: Consider the audience and purpose.



Step 3: Write two paragraphs based on the importance of your chosen topic and your personal experience with the topic.



Focusing on Process/Product Writing**Section 2: Pre-writing Activities****A. Choosing a Topic**

Choose a topic for the chapter writing assignment from the following list. You will develop your topic into a paragraph later.

1. Description: Describe a unusual dream you have had or you have heard about.
2. Expanded Definition: Define a skeptic. Give examples or an illustration.
3. Exemplification: As critical thinkers, we can be skeptical without closing our minds to new ideas. Give examples.

B. Brainstorming**C. Paraphrasing****D. Reporting and Summarizing****E. First Draft**

Expanding Your Point of View

Section 1: Peer Feedback

Now get feedback from your classmates on how clear and logical your thinking is. Get into groups of three or four. Each member of the group will read his/her paragraph to the group. Follow the "Peer Feedback Guidelines" in your Workbook.

Section 2: Revising

Paragraph Assessment Guidelines

Along with your peer feedback comments, use the following checklist to critically examine your own paragraph. Your teacher will also use it to evaluate your final draft.

Content and Ideas

- _____ There is a clear main idea.
- _____ There is enough logical support.
- _____ There are no irrelevant points.

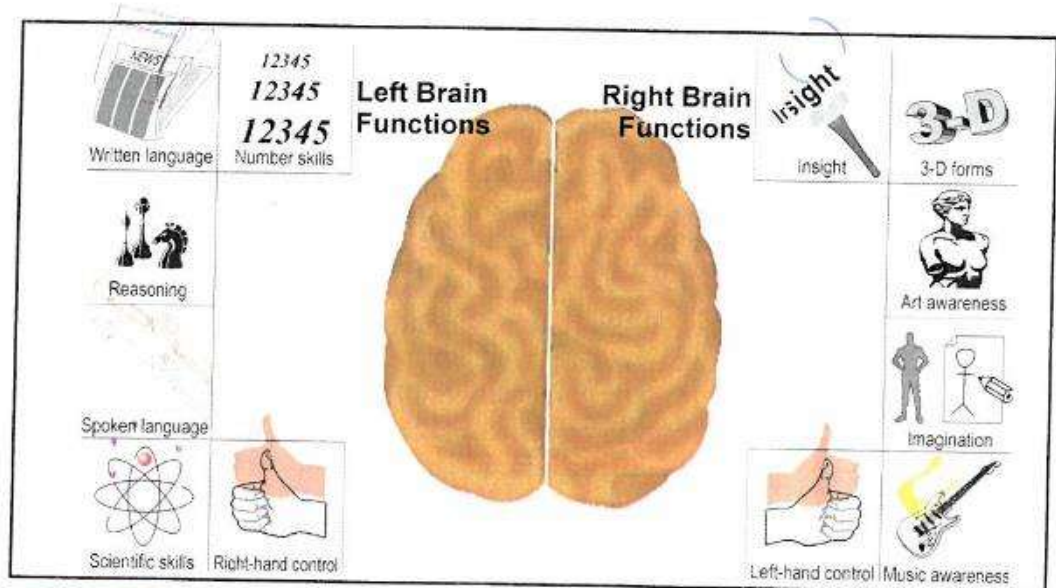
Organizational Form

- _____ There is a clear topic sentence.
- _____ It is clear which are the main supporting points and which are the secondary points.
- _____ Your sentences follow each other.

Language Conventions

- _____ There are no fragments.
- _____ There is acceptable word choice.
- _____ There is acceptable grammar.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

The World Within Us - Part F



A. Chapter Assessment

Your teacher will assess your paragraph using the "Paragraph Assessment Guidelines" found in this chapter. (Your teacher may grade your paragraph or allow you to do a further revision before assigning a grade.)

B. Sequenced Research Project

In addition to writing paragraphs and essays in each chapter of each unit, you will be required to do research on a topic of your own choice and go through several steps in writing the paper. In Units 2 and 3, each step will be a complete assignment within itself as well as part of a whole research paper. In this unit, you will cover three steps in one assignment.

Step 1: Choosing the Topic

In order to help you choose your topic, the teacher will brainstorm with the class for possibilities. Some of these possibilities will be related to the material in your textbook and/or concerns in your school or community. Skim the table of contents of your textbook for ideas. Choosing a topic involves several minor steps as well.

1. Choosing a broad topic: technology (for example)
2. Narrowing the topic: computers

Section 3: Conferencing with the Teacher

Teachers often like to check your draft before the final product is written for assessment. Your teacher may choose to pass around the classroom and check your work, take your work home in order to return it with comments and suggestions for improvement or give you time in the office to go over your writing in detail if found necessary. After the conferencing, you are to write the final product in your Workbook or on a piece of paper to hand in. At this stage, you should focus on all aspects of the writing spectrum: content, organization, vocabulary, language (grammar, sentence structure, etc.) and mechanics (spelling, capitalization, punctuation, format, neatness and legibility).

Section 4: Further Drafting



Section 5: Outlining the Final Draft

Your writing at this point has probably gone through several changes. To show the reader the main and minor ideas of your final draft, an outline can be used to serve this purpose and also to help you to see the ideas clearly for yourself. In most writing, the outline serves the purpose of helping the reader preview what is included in your text. It is usually placed at the beginning of the product and is referred to as the table of contents. Although the paragraph such as the one you have written, or the essays and project that you will be required to do throughout this textbook are not long pieces of writing, such as books, still an outline does help both the writer and the reader to view at a glance the logical order of ideas in the piece of writing concerned and helps train the writer in the logical thought processes required for each type of writing assignment.



Section 6: Publishing

Your teacher may wish to photocopy your assignments and pin them on a bulletin board with or without your name. Your work may also be published in the school's magazine or on the Internet. After all, writers write for readers and readers are always looking for interesting material to read.

Writing Skills - Paraphrasing

In Chapter 1, you practiced one strategy for paraphrasing. Another strategy is to begin the original sentence differently, changing all the nouns, verbs, adjectives, and adverbs you can as you complete the sentence. You may also make longer sentences shorter and shorter sentences longer. Do not forget to state the source.

Directions: Using this new strategy, paraphrase the following sentence taken from "The Continental Drift Theory."

The present is the key to the past.

1. If we understand ..., we ...
2. We can learn about ... from ...
3. The key to ... is found in ...
4. By understanding ..., we can ...



Language Conventions:

Comma Splices and Fused Sentences

A comma splice occurs when two complete sentences are joined together by a comma. A similar error is the joining together of two complete sentences without any type of punctuation. This type of error is called a fused sentence. Both comma splices and fused sentences are often referred to as run-together sentences. You will also find the term *run-on* used in place of the fused sentence.

Comma splice: Earthquake forecasts are an emerging, tested, scientifically sound way of making long-term statements about seismic hazards in a given region, earthquake predictions are misguided attempts by people trying to make short-term predictions.

Fused sentence: Earthquake forecasts are an emerging, tested, scientifically sound way of making long-term statements about seismic hazards in a given region earthquake predictions are misguided attempts by people trying to make short-term predictions.

What Is a Tectonic Plate?



Listening Comprehension:

Listen to the following explanation of what a tectonic plate is. Then prepare to answer the questions below. As you listen, try to figure out the meaning of these words:

massive, slab, float, crust, boundaries, drifting, clustering

ORAL DISCUSSION

1. What is feldspar?
2. What is the importance of the GEOSAT satellite measurements of oceanic plate boundaries?
3. Explain the similarity between tectonic plates and cars.

Chapter 2

The World Around Us - Part B

The Continental Drift

WARM UP

Working with a friend, try to put together the "continent jigsaw" given to you by your teacher.



What do you think about the formation of the Earth's crust? Tell the class about any theories you might have heard.



tropical: adj. living in the tropics or the hottest areas on earth.
 temperate: adj. free from very high or very low temperatures; moderate.
 debated: v. argued about.

it was physically impossible for most of these organisms to have swum or to have been transported across the ocean.

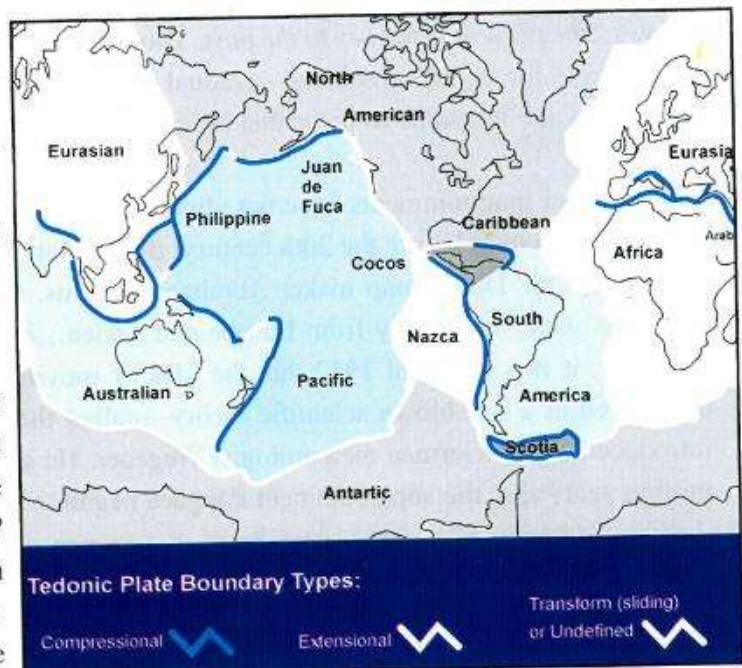
In Wegener's mind, the drifting of continents after the break-up of Pangaea explained not only the matching fossil occurrences but also the evidence of dramatic climate changes on some continents. For example, the discovery of fossils of **tropical** plants in Antarctica led to the conclusion that this frozen land previously must have been situated closer to the equator, in a more **temperate** climate.

The theory of continental drift would become the spark that ignited a new way of viewing the Earth. A weakness in Wegener's theory was that it could not satisfactorily answer the most fundamental question raised by his critics: What kind of forces could be strong enough to move such large masses of solid rock over such great distances? After Wegener's death in 1930, new evidence from ocean floor exploration and other studies led to the development of the theory of plate tectonics.

Plate tectonics has proven to be as important to the earth sciences as the discovery of the structure of the atom was to physics and chemistry. Even though the theory of plate tectonics is now widely accepted by the scientific community, aspects of the theory are still being **debated** today.

ORAL DISCUSSION

1. Why does the author refer to Earth as "the dynamic planet"?
2. Can you think of any earth sciences other than paleontology and seismology?
3. Which would you agree with more regarding the geologic changes on earth: the catastrophic principle or Hutton's principle?
4. Could you agree with Ortelius's idea if you look at the map of the world?



5. What question would you raise against the theory of plate tectonics?

READING

The Continental Drift Theory

According to the continental drift theory, the supercontinent *Pangaea* began to break up about 225-200 million years ago, eventually fragmenting into the continents as we know them today.

Plate tectonics is a relatively new scientific concept, introduced some 30 years ago, but it has revolutionized our understanding of the dynamic planet upon which we live. The theory has unified the study of the Earth by drawing together many branches of the earth sciences, from paleontology (the study of **fossils**) to seismology (the study of earthquakes). It has provided explanations to questions scientists had **speculated** upon for centuries—such as why earthquakes and volcanic eruptions occur in very specific areas around the world, and how and why great mountain ranges like the Alps and Himalayas formed.

Why is the earth so restless? What causes the ground to shake violently, volcanoes to erupt with explosive force, and great mountain ranges to rise to incredible heights? Scientists, philosophers and theologians have considered questions such as these for centuries. Until the 1700s, it was thought that a great flood played a major role in shaping the Earth's surface. Geology (the study of the Earth) was based on the belief that all earthly changes were sudden and caused by a series of **catastrophes**. However, by the mid-19th century, a new way of thinking was proposed by James Hutton, a Scottish geologist, in 1785. This principle is commonly stated as follows: *The present is the key to the past*. Those holding this viewpoint assume that the geologic forces and processes—gradual as well as catastrophic—acting on the Earth today are the same as those that have acted in the geologic past.

The belief that continents have not always been fixed in their present positions was suspected long before the 20th century; this **notion** was first suggested as early as 1596 by the Dutch map maker Abraham Ortelius. Ortelius suggested that the Americas were "torn away from Europe and Africa...by earthquakes and floods." However, it was not until 1912 that the idea of moving continents was seriously considered as a full-blown scientific theory—called the continental drift theory—introduced by the German meteorologist Wegener. He **contended** that, around 200 million years ago, the supercontinent *Pangaea* began to split apart.

Wegener's theory was based, in part, on what appeared to him to be the remarkable fit of the South American and African continents, first noted by Ortelius three centuries earlier. Wegener was also **intrigued** by the occurrences of geologic structures and of plant and animal fossils found on the **matching** coastlines of South America and Africa, now widely separated by the Atlantic Ocean. He reasoned that

READING STRATEGIES: Glossary

fossils: n. hardened parts or prints of animals or plants of long ago that have been preserved in rock, ice, etc.

speculated: v. thought about.

catastrophes: n. sudden, unexpected, terrible events that cause great suffering, misfortune, or ruin.

Earthquakes and floods are considered natural catastrophes.

notion: n. here, belief; also idea or opinion.

contended: v. claimed.

intrigued: v. greatly interested.

matching: adj. similar; fit together.

Focusing on Process/Product Writing

Writing Topics

B. Focusing on Main Ideas **Thesis Statement/Topic Sentence**

A thesis statement, usually found at the end of the introduction, facilitates the reader's understanding of what the basic content of an essay will be. A topic sentence, usually found at or near the beginning of a new paragraph, helps the reader determine the direction the author intends to take in a particular paragraph. The thesis includes the **topic**, the author's **approach** and quite often **a list** of the main supports. A topic sentence also includes the **topic** and the author's **approach**, but a list of supports is not found since the writer should focus on supporting only one main idea.



Focusing on Process/Product Writing

Section 2: Pre-writing Activities

A. Choosing a Topic

A. Choose a topic for the chapter writing assignment from the following list.
You will develop your topic into an essay later.

1. Cause/effect/example: Discuss some important effects of earthquakes. Include some recent examples from the news or other multi-media sources. (ruptures, ground shaking, land slides, tsunamis, fires, liquefaction, loss of lives and property, etc.)
2. Reason: Give reasons why one should or should not believe in Wegener's theory of continental drift.
3. Process/ description: Suggest steps or precautions that one should take in order to survive a natural disaster, such as an earthquake.

B. Brainstorming



C. Paraphrasing



D. Reporting and Summarizing



E. First Draft



Ways to Correct Run-together Sentences

Use one of the following ways.

Coordinating conjunction	(and, but, yet, so, for, or, nor)
Subordinator	(who, when, where, how, etc.)
Transition word or phrase	(however, in addition, etc.)
Period	

Directions: With a partner, find and correct any run-together sentences in the following passages about earthquake weather.

On hot days with no wind, some people will tell you that an earthquake is coming it is "earthquake weather," you had better be ready. That is simply not true.

The myth of "earthquake weather" dates back to the 4th century B.C. The Greek philosopher Aristotle believed that earthquakes were caused by winds rumbling around in huge underground caverns, restlessly searching for a way out to the surface. These winds were trapped below ground, there would be no wind above ground, it would be stiflingly hot. Clearly, Aristotle was wrong.

However, his folly took hold even today, there are people who believe in "earthquake weather." To date, however, there have been numerous studies done to try to find a link between weather and earthquakes, no such link has been found yet.

Focusing on Process/Product Writing

Section 1: Writing Topics

A. Focusing on Paragraph/Essay Formats

Writing an essay is not more difficult than writing a paragraph, just a longer task. If you can write a good paragraph, you can also write a good essay since the principles of organization for both are the same. Look at the essay as an expansion of paragraph format.

Paragraph Format

Topic sentence (Topic + Approach)

A. Support #1

B. Support #2

C. Support #3

Conclusion (Restatement of topic sentence and/or List of main supports)

Essay Format

Introductory Paragraph

General Comments

Thesis Statement (Topic + Approach + List of Supports)

Body Paragraph #1

Topic Sentence (Main Support #1)

Sub-supports (Specific Evidence)

Body Paragraph #2

Topic Sentence (Main Support #2)

Sub-supports (Specific Evidence)

Body Paragraph #3

Topic Sentence (Main Support #3)

Sub-supports (Specific Evidence)

Conclusion

Restatement of Thesis

Opinion(s)

Direction: Read the following short essay. With a classmate, fill in the outline in your Workbook.  Together decide whether or not this essay is well organized and whether or not it has sufficient content.

Academic Disciplines and Conventions

All writing follows conventions. Cookbooks, letters, novels, lists and dictionaries all depend on a specific kind of language and presentation to be comprehensible and easy to use. You can open any book with a plain cover, glance at the pages and, in most cases, know immediately what kind of book it is. We can read these conventional forms effortlessly. Academic disciplines, too, have conventional forms that their authors must use if other scholars are to understand what they are saying.

Academic disciplines often govern the kind of style and voice the author must use if the article is to be acceptable to others in the same field. For example, hard scientists are often required to emphasize the experiment and the results and downplay the contributions of the humans who conducted the experiment (i.e., "the experiment was conducted," or "the results suggest"). Social scientists may refer to those who conducted the experiments in some sections of their reports but not in others (i.e., "the data were collected," but "we note that"). Scholars in the humanities tend to avoid including themselves in their work but not always.

Academic writing also tends to have conventional formats. Scientific (social and hard) papers are generally presented with clearly labeled sections that contain a discussion of the problem, the existing literature (the articles and books people have already published on the subject), the experiment and its design, the results, and how the results fill a gap or improve understanding. On the other hand, articles written for the humanities tend to use section headings less frequently but they still generally contain three main sections: the introduction (which describes the background and the "problem"), the body (which explains the author's reasoning about and evidence for the problem), and the conclusion (which summarizes the author's answer).

Knowing that all academic articles ought to follow some fairly strict conventions in their presentation and organization is the first step in reading and understanding articles. Recognizing these conventions can also help when it comes to writing in the various disciplines. Styles in writing are somewhat like styles in clothes. You would not (presumably) wear a tuxedo to work or a ballgown to a meeting, nor would you wear jeans to a formal dance. If you recognize that academic articles are written in a particular style or fashion, you will be able to discover what the author is saying and perhaps find ways to use it in your own writing.

Chapter 2 The World Around Us - Part E

Expanding Your Point of View

Section 1: Peer Feedback

Now get feedback from your classmates on how clear and logical your thinking is. Get into small groups. Each member of the group will read his or her essay to the group. Follow the "Peer Feedback Guidelines" in the Workbook.

Section 2: Revising

Along with your peer feedback comments, use the following checklist to critically examine your own essay. Your teacher will also use it to evaluate your final draft.

Essay Assessment Guidelines

Content and Ideas

- _____ There is a clear thesis statement.
- _____ There is enough logical support.
- _____ There are no irrelevant points.

Organization and Form

- _____ There is a clear beginning, middle and end.
- _____ It is clear which are the main points and which are secondary points.

Language Conventions

- _____ There are no fragments.
- _____ There are no comma splices.
- _____ There are no run-on sentences.
- _____ There is acceptable word choice.
- _____ There is acceptable grammar.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: Conferencing with the Teacher

Section 4: Further Drafting



The World Around Us - Part F

Chapter Assessment

Your teacher will assess your essay using the "Essay Assessment Guidelines" found in this chapter.

Chapter 3

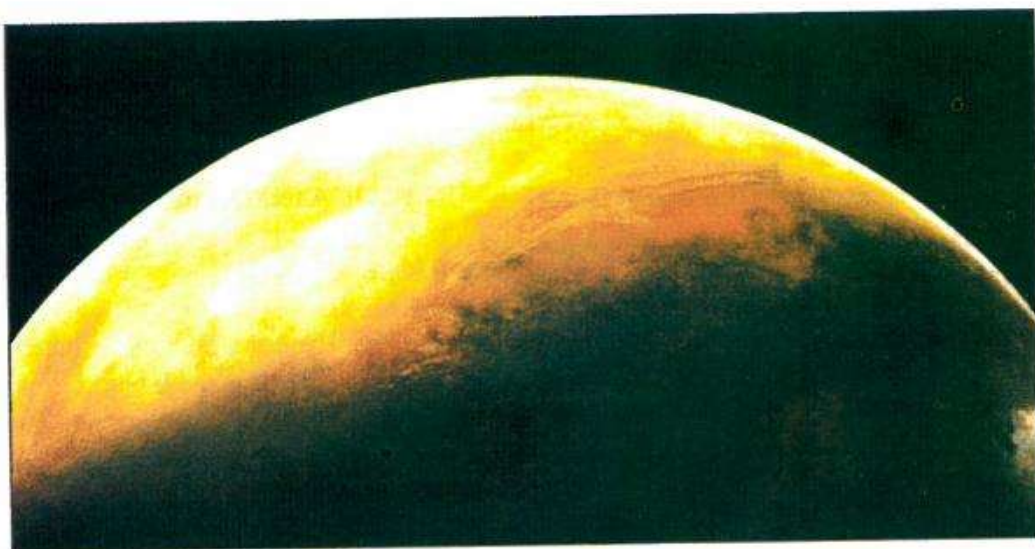
NATURAL PHENOMENA

New Worlds - Part A

Mars

WARM UP

What do you know about the planet Mars?



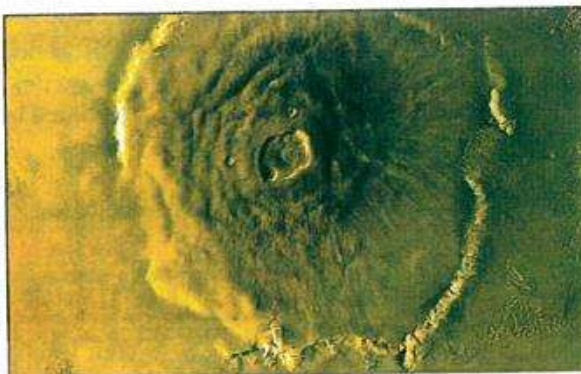
Note: the two words *optical illusion* usually occur together. When two words are most often used together, they are said to collocate very well. As you read the text below, find other pairs of words that collocate well.

READING:

The Face on Mars

(Adapted from an article by Sally Stephens, Astronomical Society of the Pacific).

The planet Mars has intrigued humans since we first began to study the sky. It is the only planet that looks red to **the naked eye**, which may explain why the ancient Greeks and Romans associated it with the bloody god of war. Spacecraft that visited Mars in the 1960s and 1970s found a freezing, arid, dead world.



READING STRATEGIES: Glossary

The naked eye: can be seen without the use of instruments or tools.

Climate on Mars

WARM UP

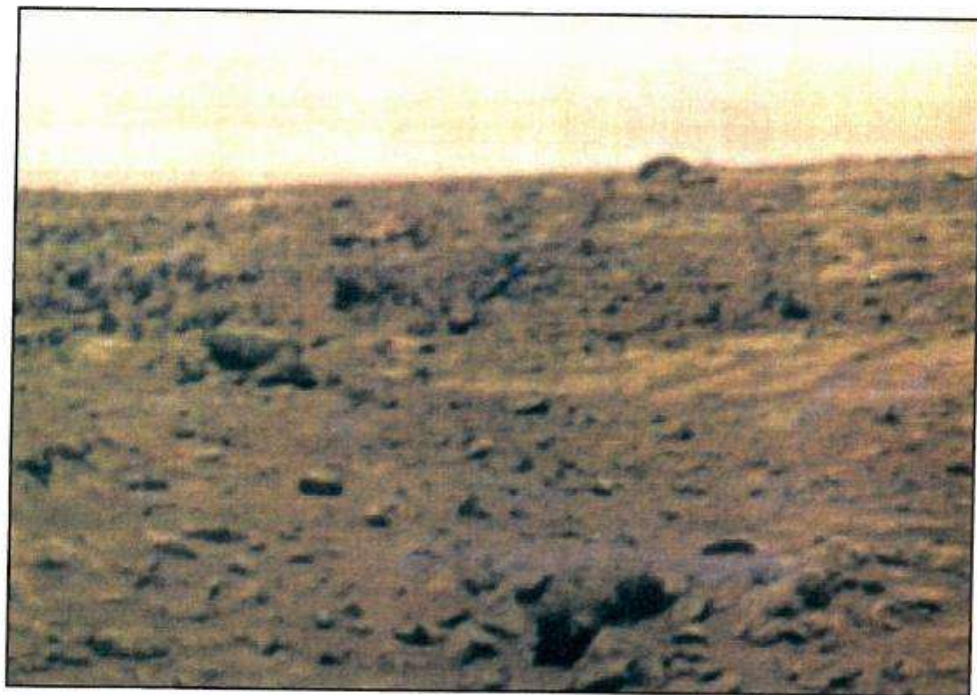
Discuss with the class the predictability or unpredictability of the weather. Note that in Lebanon, the weather is more predictable than in other regions. Nevertheless, there are times when we are surprised at an unpredicted rain storm or snow storm.

READINGS

Mars' Chaotic Climate

READING
STRATEGIES:
Glossary

plummet: v. fall
deeply and suddenly.



(1) If you think the weather on Earth is unpredictable, try living on Mars. One week, the sky is pink and cloudless, filled with windblown dust. By Martian standards, it is warm, about minus 40 degrees Fahrenheit. Then, in a matter of days, the dust is swept from the atmosphere. Temperatures **plummet** 40 degrees and brilliant water ice clouds appear against a dark blue sky.

Dramatic weather changes like these may not seem very different from a few thunderstorms passing through, but for Mars, these changes can sweep over the

the result of a gigantic construction project by intelligent beings.

Who were they and why does the Face look human? Hoagland has several theories. Perhaps evolution worked the same way on Mars as it did on Earth and so the Martians looked human. Or maybe a previously unknown, technologically advanced civilization from Earth's distant past traveled to Mars (or, alternatively, an advanced Martian civilization came to Earth long ago, and we look like them). Or perhaps the face was built by some sophisticated extraterrestrials from beyond the solar system as a signal for us.

Is the Face real? Although the rock outcropping on Mars suggests a human face, that does not mean it is a monument deliberately constructed to look like a face. According to psychologists, the human visual system is organized to look for familiar features in random patterns. For example, people all over the world see the face of a man when they look at the Moon; others see other shapes. We've all seen people in the news or on TV who have found vegetables that look like human faces. No one thinks that any of these were deliberately created, so why should the Face on Mars be different?

ORAL DISCUSSION

1. What myths or fictitious stories have you heard about Mars?
2. Explain the last sentence in paragraph 2: "Such extraordinary claims require extraordinary proof."
3. Have you ever experienced seeing "apparently familiar shapes in complex landscapes"? Tell the class.
4. Look on the left and right sides of the maple leaf on the Canadian flag. What do you see?



At home:

On a clear night when the Moon is full, look at it and try to sketch on a piece of paper what you see.

Hang your sketch on one of the walls in your classroom (or a board if there is one). Compare and contrast the various sketches done by your classmates.

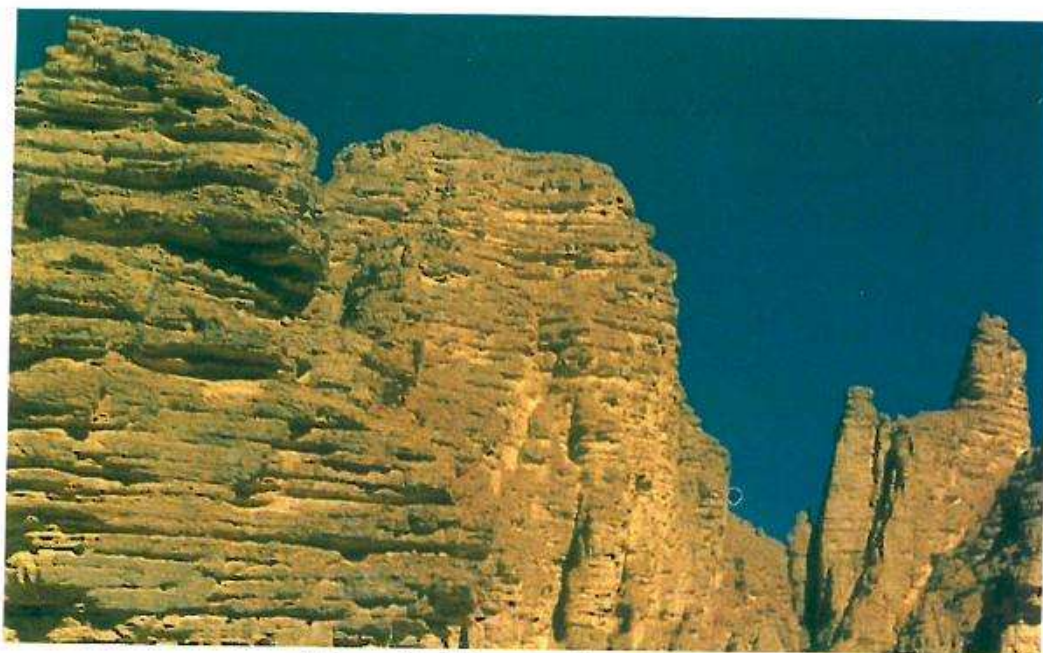
Which one do you think is the most imaginative? Which one is the closest to reality?



Nevertheless, the idea of life on Mars, and, in particular, intelligent life on Mars, **persists** in the public mind, despite the weight of scientific evidence against it.

This myth has surfaced again in the so-called "Face on Mars," a rock **outcropping** that looks like a human face when lit from the side. Astronomers see it as a mere **optical illusion**, proof of the power of human imagination. A few individuals, however, see it as proof of an ancient Mars civilization. Some of them even go so far as to accuse NASA of deliberately destroying the billion-dollar *Mars Observer* spacecraft to keep from having to admit that the Face is "real." Such extraordinary claims require extraordinary proof, which is **sorely** lacking in this case.

In 1976, two unmanned *Viking* spacecraft went into orbit around Mars. Each sent a lander down to the surface, while the orbiters radioed back over 60,000 photographs of the Martian surface. Not long after the orbiters began their mission, *Viking* scientists noticed a picture of a mile-long hill or **mesa** in the Cydonia region of Mars that looked like a face. They released the image to the press as a joke, an example of our tendency to see apparently familiar shapes in complex landscapes. The researchers and the press recognized the Face as an unusual rock formation created by weathering. The dramatic low angle of late-afternoon lighting in the photo made the outcropping look like a face.



A Strange Rock Formation Found in Lebanon

Three years later, two computer scientists with no particular **expertise** in Martian geology came across the images while going through the Viking archives. They experimented with some **image-enhancement** programs and concluded that the Face did not occur naturally. They also noticed several "pyramids" near the Face and published a book calling attention to the structures.

In the 1980s, Richard Hoagland, a science journalist, took up the cause of the Face on Mars in several books and numerous radio and TV appearances. Hoagland sees evidence of a ruined "city" and a "fort." To Hoagland, this region of Mars is

persists: v. continue in spite of opposition.

outcropping: v. rock which stands up out of the ground.

optical illusion: something that deceives the eye.

sorely: adv. severely; very.

mesa: v. a flat-topped formation with steep rocky walls.

expertise: v. skill in a particular field.

image-enhancement: improvement by using technical devices such as computers.

Guideline 6: Although they are often plural in meaning, most indefinite pronouns, such as *everybody*, *somebody* and *everything*, etc., take a singular verb.

Everybody **is** entitled to his own opinion about the predictability or unpredictability of the weather.

Guideline 7: *Neither* and *either* used as pronouns take singular verbs.

Neither of the theories **has** been proven right.

Guideline 8: Group nouns, including *family*, *class*, *committee*, *government*, *team*, *audience* and *group*, usually take a singular verb.

Our class **was** uncertain about whether or not it is easy to predict the weather in Lebanon.

Guideline 9: Some nouns ending in -s take singular verbs, such as *physics*, *mathematics*, *series* and *news*.

The latest news about new space explorations **is** always interesting to an inquiring mind.

Language Conventions**Subject-verb Agreement**

Directions: Read the following passage and correct any subject-verb agreement errors. Work in pairs.

Mars, like Mercury and the Moon, appear to lack active plate tectonics at present. There is no evidence of folded mountains, which is so common on Earth. With no lateral plate motion, hot-spots under the crust stays in a fixed position. This, along with lower surface gravity, probably account for its enormous volcanoes. However, there are no evidence of current volcanic activity. Yet, there is some new information that Mars may have had tectonic activity in its early history, making comparisons with the Earth all the more interesting.

Writing Skills - Paraphrasing

In Chapters 1 and 2, you practiced two different strategies for paraphrasing. Choose the strategy which worked better for you and do some further practice in your Workbook.



Section 2: Conventions – Subject-verb Agreement

To make subjects and verbs agree in your writing, follow these guidelines.

Guideline 1: Subjects connected by *and* are plural.

The "Face on Mars" and several "pyramids" near the Face **have** been subjects of great debate.

Guideline 2: Phrases that come between the subject and the verb do not affect the verb.

The idea of intelligent life on Mars, fed by science fiction stories, still **persists**.

Guideline 3: The verb agrees with the noun that follows the verb in the *there* structure.

There **is** no evidence to suggest that the Face on Mars was deliberately constructed to look like a face.

Guideline 4: In adjective clauses, *who*, *which* and *that* when used as subjects agree with their reference.

Astronomers who **see** the Face as a mere optical illusion do not see it as proof of an ancient civilization on Mars.

Guideline 5: The verb agrees with the second subject when using *either...or* and *neither...nor*.

Neither the weather on Earth nor the weather on Mars **is** always predictable.

Either the Face or the "pyramids" **are** the subject for the next class discussion.

snapped: v. here, taken
(as photographs);
note: a photograph is
also called a snapshot;
abrupt: sudden and
unexpected;
blanketed: covered all
over.

entire planet every week. It appears that Mars' weather is more chaotic and unpredictable than scientists first thought. Observations by the Hubble Space Telescope show that the atmosphere of Mars is more complex and variable than the picture revealed by the *Viking* and *Mariner 9* orbiters. Images **snapped** by the orbiters revealed huge dust storms spreading throughout the entire atmosphere when Mars was closest to the sun. However, information captured by Hubble shows that Mars is more often cloudy than dusty, experiencing **abrupt** swings between dusty and hot and cloudy and cold. A state of emergency would be declared on Earth if an ice or dust storm **blanketed** the entire planet.

These shifts in climate are driven by three important factors: Mars' thin atmosphere, its elliptical orbit around the sun, and strong climatic interactions between dust and water ice clouds in the atmosphere. Mars' atmosphere is so thin that it weighs less than 1 percent of Earth's atmosphere. Because Mars' atmosphere is so paper-thin and there are no oceans to store up heat from the sun, the planet's temperatures respond more quickly and intensely to surface changes and atmospheric heating by the sun. Besides, because Mars' distance from the sun varies by 20 percent in its orbit around the sun every two years, there are much larger annual changes in sunlight falling on Mars than on Earth.

ORAL DISCUSSION

1. How many degrees Celsius is minus 40 degrees Fahrenheit?
2. Why are we so interested in Mars?
3. Can you explain the significance of the ozone layer around Earth's atmosphere from the description of the climate on Mars?

Focusing on Process/Product Writing

Section 1: Writing Topics

Focusing on Patterns of Organization

Understanding a text goes beyond identifying its main idea. You must also be able to recognize the way in which pieces of information – that is, the text's supporting ideas – are related. How do English writers order these supporting ideas? Basically, the following major thought relationships or patterns of organization are usually used when writing for the sciences.

1. Definition/Extended definition
2. Process
3. Description
4. Cause/effect
5. Reason
6. Exemplification
7. Classification

These relationships are seldom used exclusively, but rather are often combined in various ways. Furthermore, each may be mentioned only briefly in a text or one might even become the general focus of the text.

Directions: In the following excerpt, find examples of as many different patterns of organization as you can, using the list above.

Disasters – floods, droughts, epidemics, and war – can cause drastic changes in one's life. Accidents happen. People we love die. Government policy may be changed or the bottom might fall out of the stock market, putting people out of work, threatening the most carefully planned lives.

How can we adapt to these changes? Adjustment is an ever-changing balance between our needs and desires, on one hand, and between the demands and restraints of the environment, on the other hand. Like adaptation in biology, adjustment implies changing in order to survive.

A young mother talks to her neighbors about how well her son has "adjusted" to his first day at school. A 70-year-old engineer may speak about how hard it was to "adjust" to retirement. A couple who have been childless for many years might find it hard to "adjust" to the demands of a baby. In each of these examples, adjustment refers to the individual's successful or unsuccessful adaptation to change.

People adjust to change in a variety of ways. An earthquake victim who has lost all his possessions or family members, for example, may have to move to a new location, ask for aid, or simply collapse and end up in a hospital. All of these reactions are ways of "adjusting."

Focusing on Process/Product Writing

Section 2: Pre-writing Activities

A. Choosing a Topic

Choose a topic for the chapter writing assignment from the following list. You will develop your topic into an essay later.

1. Extended definition / exemplification / reason : What is an optical illusion?
Give examples. Include an example from the reading texts in this chapter. Expand further by including reasons for these optical illusion. (the Face, the Pyramids on Mars, a mirage in the desert, the Canadian flag, etc.)
2. Description: The Face on Mars is a naturally occurring rock outcropping that just happens to resemble a face when light strikes it in a certain direction. Where else have you seen a naturally occurring object that looks like something it is not? Describe it in as much detail as possible.
3. Additional topic: Choose a topic that you and your teacher agree on related to natural phenomena.

B. Brainstorming



C. Paraphrasing



D. Reporting and Summarizing



E. First Draft



Expanding Your Point of View

Section 1: Peer Feedback

Now get feedback from one of your peers. Read your essays to each other. Follow the "Peer Feedback Guidelines" in your Workbook.

Section 2: Revising

Along with your peer feedback comments, use the following checklist to critically examine your own essay. Your teacher will also use it to evaluate your final draft.

Essay Assessment Guidelines

Content and Ideas

- _____ There is a clear thesis statement.
- _____ There is enough logical support.
- _____ There are no irrelevant points.

Organization and Form

- _____ There is a clear beginning, middle, and end.
- _____ It is clear which are the main supporting points and which are secondary points.

Language Conventions

- _____ There are no fragments.
- _____ There are no comma splices.
- _____ There are no run-on sentences.
- _____ There are no problems with subject-verb agreement.
- _____ There is acceptable word choice.
- _____ There is acceptable grammar.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: Conferencing with the Teacher

Section 4: Further Drafting



New Worlds - Part F

Chapter Assessment

Your teacher will access your essay using the "Essay Assessment Guideline" found in this chapter.

Chapter 1

TECHNOLOGY

The World Within Us - Part A

The Mind Candy Kafe

READING STRATEGIES: Glossary

Shift: n. change from one situation to another.

Media spots: advertisements.

WARM UP

What are the sources from which you get information? List them on the board.

Some words have the same forms as nouns and verbs. Two words are like this in the glossary: *shift* and *display*. To be able to tell which form is being used, the reader has to look at the context in which the words are used. The article *the* preceding *shift* indicates a noun. *Displays* is preceded by an adjective, *dazzling*, which also indicates that it is a noun.



READING

Read this paragraph, then do the group activity **BEFORE** you move on to the rest of the text.

Along with the many new information products flooding the marketplace, we can see a transformation in the way young people and adults come to know and understand their world. With the **shift** toward electronic media and information, the challenge of knowing and comprehending is complicated by a movement toward superficial presentations. Deep thinking, deep reading and deep commentary are replaced in many situations by *sound bites*, *mind bites*, *eye candy*, and *mind candy*.

In groups, try to explain what you think the words in italics might refer to. After you come to an agreement about the possible meanings, share them with the rest of the class.

Now continue reading.

Just what is meant by *sound bites*, *mind bites*, *eye candy* and *mind candy*?

Sound bites are brief (15 to 30 second) **media spots**, often commissioned by politicians or groups, which attempt to state a position or explain an argument on a

complex subject or issue. Since time is scarce and expensive, sound bites rely on key words, phrases, and images to make their point. Packaging is more important than content. The medium is more important than the message. Ideas are condensed in simple, highly **impact** terms.

Also in groups, discuss the following points:

1. Agree on a suitable example of a sound bite. Write it on a piece of paper. Submit it to your teacher.
2. Now, try to figure out what the meaning of the other italicized words in the first paragraph are.

Here is the rest of the text.

Mind bites are related to sound bites but more **apt** to appear in newspapers, magazines and other print media, as they attempt **to keep pace with** their electronic relatives. We see some newspapers emphasizing photo opportunities and replacing serious news with **scandals** and stories based on **hearsay**. Complex issues that relate to the society as a whole find themselves summarized in a few paragraphs.

Eye candy is a term first applied to the attractive format of TV programs and channels. We have come to expect **dazzling displays** of graphics with fast moving video and musical elements that grab the viewer's attention forcefully and intensely. Once again, we see that packaging may be more important than content.

Mind candy is a more recent term arising out of the Internet. ...

Also in groups, finish the paragraph above. Present your work to the teacher.

ORAL DISCUSSION

1. After the teacher reads each group's definition, discuss the definition and rate it from 1 to 5, where 1 is off the subject and 5 is almost similar to the definition in the text.
2. Similarly, listen to your teacher reading the definition of *mind candy*. Rate the definitions. Which group scored best?

Assignment

Working in groups, try to agree on "antidotes" to the trend for acquiring knowledge superficially as presented in the text above. Come up with practical suggestions as to how a person can immunize himself or herself from being supplied with sound bites, mind bites, eye candy and mind candy. Be ready to present your list and defend it to the rest of the class during the next English period.

Impact: n. effect.

Apt: adj. suitable.

Keep pace with: v. to stay alongside.

Scandal: n. an improper action or state.

Hearsay: n. rumor.

Dazzling: adj. splendid.

Display: n. show.

THE WORLD OF HUMAN INTERVENTION



Unit II: Technology

Unit Opener

"For I dipped into the future, far as human eye could see;
saw the vision of the world, and all the wonder that could be."
Alfred, Lord Tennyson, 1842

"There are things that are known and things that are unknown; in
between is exploration."

Anonymous

Shifts in Tone

Colloquial language and slang should be avoided when writing formally. For example, *one can make bucks in the information industry*. *Bucks* should be changed to *money*.



Electronic Robbery

WARM UP

Why do you think this chapter is entitled "Electronic Robbery"? What do you think the term means?

READING

The New Plagiarism

There are complaints that new technologies have made it all too easy for students to gather the ideas of others and present them as their own. In order to avoid plagiarism, read the following **antidotes**.

Antidote 1:

There are three different levels of research. Students should avoid the first two levels and stick to the third level.

Level One Research: Just the facts

Here, the student gathers basic facts and information about a topic, such as a country or a battle or a famous personality. Little thinking is required.

Level Two Research: Other people's ideas

Even though students might engage in considering important questions such as the causes of air pollution or overpopulation, it is sufficient for the student to gather "conventional thinking" and the best ideas of others.

Level Three Research: In my humble opinion

When fresh thinking is required, plagiarism is cut to the minimum. If students cannot find the answers but must make the answers, they are less apt to pass off others' ideas as their own. Students should start asking questions of their own: How can peace be restored to East Europe?

Antidote 2:

Students should ignore "trivial pursuits." This is the acquisition of insignificant pieces of information used as evidence. Rather, students should focus on critical analysis of information: why, how, which is best. Instead of asking: Why have events turned out that way? students can ask: Why haven't various outcomes

READING STRATEGIES: Glossary

Antidote: n. something that counteracts an undesirable situation.

occurred? Hypothetical situations lend themselves more to critical analysis. A topic that is more suitable for research than a particular city is: Why is City X an appropriate choice for the Olympic Games?

Antidote 3:

Students should try to ask and answer "essential" questions, questions that touch upon basic human issues.

Antidote 4:

Students should be able to synthesize situations, adding their own "flavor" by expressing their own opinion. Students must first be able to identify the **underlying** thoughts and then construct or deconstruct the original until it becomes their own.

Underlying: adj. basic.

Cite: v. to name.

Antidote 5:

Storing and organizing information help students in recognizing the source of their ideas. When students extract, paraphrase, summarize and extend information they have found, they should be able to **cite** the source of their information and ideas. A record should be kept by using a standard format such as the one below:

Source: (author, title, and date)
Subject:
Keywords:
Information:

The use of index cards facilitates the organization of data collected from various sources.

Antidote 6:

Students should use "green ink." They should differentiate between the ideas they have collected from others and those ideas which have emerged in reaction to the ideas of others. Students can use black text to signify the ideas of others and green text to signify fresh thinking.

Oral Discussion

1. How close were you in your assignment answers to those presented in this lesson?
2. Do you agree that the antidotes suggested in this lesson are effective in minimizing plagiarism?
3. What, in your opinion, is the most effective of the 6 antidotes? Justify your choice.

Focusing on Process/Product Writing

Section 1: Focusing on Supporting Evidence and the Unity of Ideas

No matter what the writer's purpose is or who the audience is, a writer must learn to support his/her ideas in order to

1. gain credibility,
2. clarify ideas, and
3. be more convincing.

There are several ways to support ideas. Although the audience and purpose most certainly influence your choice of support, the following are some typical types of support writers often choose.

1. Facts
2. Statistics
3. Expert testimony
4. Personal experience
5. Examples of different types – specific, extended, and hypothetical

A writer can also choose from a wide variety of sources for supporting material. The following are some areas that you may wish to search for evidence.

1. Primary sources – the actual text
2. Secondary sources – what others have written about the primary source in the form of critiques and/or analyses
3. Multi-media resources – the Internet, magazines, television, radio, etc.

Whether the supporting evidence is based on the writer's own ideas or taken from other sources, all ideas and supporting evidence must be relevant to the topic chosen. Otherwise, the reader may misunderstand the writer's purpose. Another important point is to make the supports as specific as possible. For example, instead of simply saying *questioning is the basis for information literacy*, the writer might specify. *Questions enable us to search for pertinent information, convert data into information and information into insight, interpret information, and support synthesis and the development of new understanding.* Which one is more specific and, therefore, more enlightening for the reader?

If the writer uses information from outside sources whether in the form of a summary, paraphrase or direct quotation, credit must be given to the source by mentioning the author's name, title of the work, and place and date of publication.

Writing Skills: Summarizing Passages

One of the most useful writing skills is the ability to write a summary. A summary is a short version of a text. Its purpose is to give the reader a clear idea of the most important information in a text. A summary is always quite a bit shorter than the original text, generally ranging from one sentence to about one-fourth the length of the original. A final characteristic of a summary is that it is written in your own words rather than copied from the original.

Once you have read – and reread – the text in order to understand what the author is trying to say, you should decide on how much of this information will be put into your summary. Although your choice will depend to some extent on the length and purpose of your summary, some general guidelines can be followed. A good summary usually includes:

1. the author's main idea and the source in the first sentence.
2. the author's most important supporting evidence, not detailed.
3. your own wording. However, a phrase in the original which is especially striking, interesting or controversial may be quoted.
4. a reminder to the reader that you are summarizing someone else's idea.
5. an indication of the author's attitude towards the subject matter, his or her opinion, not your own.



Language Conventions:

Achieving Consistency in Tense, Person, Number, and Tone

Shifts in Tense

Inconsistent shifts in tense can confuse a reader. For example, *along with the many new information products flooding the marketplace, we had witnessed a transformation of the way young people and adults come to know and understand their world.* Since this statement is factual, it would be appropriate to use *are witnessing/have witnessed* instead of *had witnessed*. Consistent use of tenses makes your writing easier to read and more effective.



Shifts in Person and Number

Illogical or unnecessary shifts in person or number can also confuse your reader. For example, *if students always surf on the Internet, will he ever learn to think for themselves?* If you change *he* to *they*, the question becomes logical.



Essay Assessment Guidelines

Content and Ideas

- _____ There is a clear thesis statement.
- _____ There is enough logical support.
- _____ There are no irrelevant points.

Organization and Form

- _____ There is a clear beginning, middle and end.
- _____ It is clear which are the main supporting points and which are secondary points.

Language Conventions

- _____ There are no fragments.
- _____ There are no comma splices.
- _____ There are no run-on sentences.
- _____ There are no problems with subject-verb agreement.
- _____ There is consistency in tense, person, number, and tone.
- _____ There is acceptable word choice.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: Conferencing with the Teacher.

Section 4: Further Drafting



If this is not done, the writer plagiarizes. Giving credit to borrowed supporting material is referred to as citing or documenting your material. One of the easiest ways to cite is to place a citation within the text and give the full reference at the end of the writing. For example, one way to acknowledge the source would be as follows:

What can be termed new plagiarism is an issue of great importance today. Electronic texts have spawned a new strain of student copying. "We see many young people scooping up and saving dozens of pages of information without necessarily bothering to read, digest, challenge or synthesize." (McKenzie, 1998, p.8)

This quotation has been carefully selected to support a point. The format for this citation is written in the APA style. Another common style is the MLA. In this case, the citation would be (McKenzie 8). Is there a difference in format? For papers written for the sciences, the APA is the most common format used.



The World Within Us - Part E

Expanding Your Point of View

Section 1: Peer Feedback

Now get feedback from one of your peers. Read your paragraphs or essays to each other. Follow the "Peer Feedback Guideline" in your Workbook.

Section 2: Revising

Along with your peer feedback comments, use the following checklist to critically examine your own paragraph or essay. Your teacher will also use it to evaluate your final draft.

Paragraph Assessment Guidelines

Content and Ideas

- _____ There is a clear main idea.
- _____ There is enough logical support.
- _____ There are no irrelevant points.

Organizational Form

- _____ There is a clear topic sentence.
- _____ It is clear which are the main supporting points and which are the secondary points.
- _____ Your sentences follow each other.

Language Conventions

- _____ There are no fragments.
- _____ There are no comma splices.
- _____ There are no run-on sentences.
- _____ There are no problems with subject-verb agreement.
- _____ There is consistency in tense, person, number and tone.
- _____ There is acceptable word choice.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Focusing on Process/Product Writing*Section 2: Pre-writing Activities***A. Choosing a Topic**

Choose a topic for the chapter writing assignment from the following list and write a paragraph or an essay depending upon your teacher's instructions. Focus on the main organizational types which are indicated below.

1. Enumeration: It is often said today that we live in a world where there is more and more information, but less and less meaning. You have been asked in this chapter to list ways in which a person can immunize himself or herself against sound bites, mind bites, eye candy and mind candy. Discuss what can be done to oppose this trend towards blindly accepting plastic, superficial knowledge.
2. Expanded definition: Define mind candy. Use information collected from the class discussion. Include examples.
3. Additional topic: Choose a topic that you and your teacher agree on related to information technology.

B. Brainstorming



C. Paraphrasing



D. Reporting and Summarizing



E. First Draft



A. Chapter Assessment

Your teacher will assess your paragraph or essay using the guidelines found in this chapter.

B. Sequenced Research Project

In this unit, you will write a survey report in three steps, one for each chapter.

Step1: Choose and analyze the sample.

Step2: Analyze and arrange the data.

Step3: Write the report.

Step 1: Choose the sample.

You are to question two types of people about your topic. The first type or sampling would be people who have no reason to be interested in or knowledgeable about your topic. The second type should be interested in your topic. Choose people who have been affected by the issue or people who know a great deal about your topic. In order to get people to cooperate with you, here are some friendly tips.

1. Be sure the person you interview has time to talk to you. This may mean making an appointment ahead of time.
2. Explain what the survey is about and why you are doing it.
3. Tell your participant that all answers will remain confidential.
4. Have questions prepared ahead of time and take notes. A tape recorder may also be used to collect information.

When you have spoken to as many people as you can (at least a sampling of ten), analyze your sample by answering the following questions.

Who did you question?

What characteristics do they have in common (age, occupation, etc.)?

Why are they interested in your topic?

What kind of knowledge do they have about your topic?

How does your topic affect them personally?

You will analyze and arrange this data in Step 2 at the end of the next chapter.

aged five to fourteen. New research on using plant expression systems will provide low cost vaccines to the developing world.

Outcome: n. result.

Bananas for Vaccine Production

Along these lines, research is concentrating on gene expression in banana fruit during development and ripening. The desired **outcome** is a food that can be eaten uncooked (to prevent vaccine denaturation) by children and that can be grown in many developing countries.

ORAL DISCUSSION

1. Why is banana fruit—rather than other fruits—being researched on?
2. What other fruits would have also been suitable?
3. If you were a Chinese official, which food would you ask Chinese scientists to develop for an "edible vaccine"?

Biotechnology

WARM UP

The previous lesson introduced telemedicine; this lesson will introduce biotechnology. Both of these topics are based on advanced technology that is rapidly spreading in our world. Can you think of other practices that are similarly spreading?

READING

Biotechnology refers to a broad set of techniques which allow the genetic manipulation of organisms. It has been used to develop improved crops and many pharmaceutical products. Biotechnology is a comparatively new area of science and commerce.

READING STRATEGIES: Glossary

Edible: adj. can be eaten.
Elicit: v. bring out.
Life threatening: can cause death.

Edible Vaccines

During the last ten years, biotechnology has provided new strategies for designing vaccines. These advances are based on techniques that allow the isolation of genes from viruses or bacteria that cause human or animal disease. One technique is to produce vaccines from yeast cells grown in fermentation vats. But such techniques are still too costly to be distributed widely in all regions of the world. Research is being made on crops that will produce "**edible** vaccines," which will **elicit** an immune response when the food is eaten.

Vaccines against Diarrhea

It was estimated in 1992 that the lives of 3 to 5 million children could be saved each year if vaccines were available to control or prevent common diseases. Children in poor countries often suffer from diseases that are not usually **life threatening** in developed countries, for example, diarrhea. In the developing world, one-third of the deaths from diarrhea are in children under five years of age and another one-quarter are in children

Writing Skills - More Summarizing

In Chapter 1, you practiced summarizing texts which have three parts, like an article with an introduction, body paragraph(s), and a conclusion. You will now practice a strategy for summarizing smaller bits of information from texts, rather than summarizing an entire article, book, etc. As with longer texts, your aim is to capture the essence of what the author says, but in fewer words.

Using the following passages on telemedicine and biotechnology, follow this strategy:

- A. Read the selected passage several times.
- B. Underline key words or phrases that remind you of the author's main point(s), not the details.
- C. Expand some of these key words or phrases into one or two sentences.
- D. Try to use 3rd person pronouns (he, she, it, they) even if the text is written in 1st person.

1. Home health care is another booming area of telemedicine. A program in Japan has home-bound patients communicating daily with a doctor, nurse or physical therapist. Telemedicine does not have to be expensive. Many projects are providing valuable services to those with no access to health care.

Summary: _____

2. During the last ten years, biotechnology has provided new strategies for designing vaccines. These advances are based on techniques that allow the isolation of genes from viruses or bacteria that cause human or animal disease. One technique is to produce vaccines from yeast cells grown in fermentation vats. But such techniques are still too costly to be distributed widely in all regions of the world. Therefore, research is being done on crops that will produce "edible vaccines," which will elicit an immune response when the food is eaten.

Summary: _____

Chapter 2

TECHNOLOGY

The World Around Us - Part A

Telemedicine



WARM UP

What do you think the term *telemedicine* means? Can you tell by dividing it into two parts: *tele-* and *medicine*?

The suffix *-ology* is derived from Greek to mean "the study of." Three words having this suffix are listed to the side in the glossary. What other words do you know that end with the suffix *-ology*? (*geology, gemology, theology, etc.*) Also, *-ics* refers to certain studies, such as *pediatrics* and *obstetrics*.

The text below uses the prefix *tele-* which means "at or over a distance." Pick out these words and explain how *tele-* applies.

READING

READING STRATEGIES: Glossary

Forward: v. transmit.

Telemedicine has been defined as the use of telecommunications to provide medical information and services. It may be as simple as two doctors discussing a case over the telephone, or as sophisticated as using satellite technology to broadcast a consultation between hospitals in two countries using videoconferencing equipment. The first is used daily by most health professionals, and the latter is used largely by medical centers.

Two different kinds of technology make up most of the telemedicine applications in use today. The first, called store and **forward**, is used for transferring digital images from one location to another. A digital image is taken using a digital camera, (stored) and then sent (forwarded) to another location. This is typically

used for non-emergency situations, when a **diagnosis** or consultation may be made in the next 24 to 48 hours and sent back.

The image may be transferred within a building, between two buildings in the same city, or from one location to another anywhere in the world. Teleradiology, the sending of x-rays, CT scans, or MRIs (store-and-forward images) is the most common application of telemedicine in use today. There are hundreds of medical centers, clinics, and individual physicians who use some form of teleradiology. Many radiologists are installing teleradiology equipment in their homes, so they can have images sent directly to them for diagnosis, instead of making an off-hours trip to a hospital or clinic.

Telepathology is another common use of this technology. Images of **pathology** slides may be sent from one location to another for diagnostic consultation. Dermatological images are also a **natural** for store and forward technology. Digital images may be taken of skin diseases and sent to a dermatologist for analysis.

The other widely used technology, two-way interactive television (IATV), is used when a consultation between the patient, primary care provider and specialist is necessary. Videoconferencing equipment at both locations allows a "real-time" consultation to take place. Almost all specialties of medicine have been found **to be conducive to** this kind of consultation including psychiatry, internal medicine, **rehabilitation, cardiology, pediatrics, obstetrics and gynecology** among others.

Home health care is another booming area of telemedicine. A program in Japan has home-bound patients communicating daily with a doctor, nurse or physical therapist. Telemedicine does not have to be expensive. Many projects are providing valuable services to those with no access to health care.

The use of telemedicine can provide several advantages. It can make specialty care more accessible. Video consultations from a rural clinic to a specialist can reduce travel for patients. Videoconferencing also opens up new possibilities for educating doctors who are unable to join conferences far away.

There are still several barriers and disadvantages to the practice of telemedicine.

If you are interested in more information about telemedicine, this is a helpful address on the Internet: <http://www.telemed.org>

ORAL DISCUSSION

1. How do you think telemedicine can improve our capacity for handling serious medical situations?
2. What is a "primary care provider"?
3. What is the advantage of "real-time consultation"?
4. Note that the disadvantages of telemedicine are not given in the article. Can you think of any?

Diagnosis: n. the act of identifying a disease from its signs.
Pathology: n. the study of the essential nature of diseases.
Natural: n. here: informal to mean a very suitable candidate.
Real-time: a term that refers to two things happening at the same time in real life.
Be conducive to: tending to be assisted by.
Rehabilitation: n. restoration to a healthy condition
Cardiology: n. the study of the heart and its action and diseases.
Pediatrics: n. a branch of medicine dealing with the development, care, and diseases of children.
Obstetrics: n. a branch of medicine dealing with birth and conditions that precede and follow it.
Gynecology: n. a branch of medicine dealing with the diseases and routine physical care of the reproductive system of women.

Chapter Assessment

Your teacher will assess your essay using the guidelines found in this chapter.

Sequenced Research**Writing a Report**

Step 2: Analyzing and Arranging the Data

Look at your collected data and arrange your data.



Section 2: Pre-writing Activities

A. Choosing a Topic

Choose a topic for the chapter writing assignment from the following list and write an essay. Focus on the organizational types which are indicated below.

1. Enumeration/Description: What steps should be taken to improve health services in your area?
2. Reason/Exemplification: Give reasons why Lebanon should emphasize or de-emphasize technology as we move towards the future. Include examples.
4. Student generated topics: If you and your teacher agree on additional topics related to technology, list them and choose the one that interests you the most.

B.. Brainstorming



C. Paraphrasing



D. Reporting and Summarizing



E. First Draft



Expanding Your Point of View**Section 1: Peer Feedback**

Now get feedback from one of your peers. Read your essays to each other. Follow the "Peer Feedback Guidelines" in your Workbook.

Section 2: Revising

Along with your peer feedback comments, use the following checklist to critically examine your own essay. Your teacher will also use it to evaluate your final draft.

Essay Assessment Guidelines**Content and Ideas**

- _____ There is a clear main idea.
- _____ There is enough logical support.
- _____ There are no irrelevant points.

Organizational Form

- _____ There is a clear topic sentence.
- _____ It is clear which are the main supporting points and which are the secondary points.

Language Conventions

- _____ There are no fragments, comma splices, or run-on sentences.
- _____ There are no problems with subject-verb agreement.
- _____ There is consistency in tense, person, number, and tone.
- _____ There is sentence variety.
- _____ There is acceptable word choice.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: Conferencing with the Teacher**Section 4: Further Drafting**

In groups of three or four, exchange your summaries. Did your peers choose the same key words as you did? Are your summaries similar?



Language Conventions: Sentence Variety

A good writer strives to add interest to his writing by using a variety of sentence types: simple, compound, and complex. If a writer uses one kind over and over, it is certain that he will bore his reader. Read the following text which lacks sentence variety.

Telemedicine allows medical practitioners to help their patients in a number of innovative ways. A surgeon can now operate on a patient in remote locations. Telesurgery is one example. It enables a surgeon to assist medics on a battlefield. Telemedicine is also being used for local surgery. An example is in endoscopic surgery. Surgeons manipulate instruments by viewing a television monitor. They manipulate a tool. It is inserted through a tube. It is inserted into a patient. Even dentists are using 3-D eyeglasses. They divert a patient's attention while sitting in the chair. They reduce anxiety or stress.

In order to make the text above more interesting, the writer could (1) expand the simple sentences by adding or using various types of phrases, (2) use a compound sentence if the ideas are of equal importance to him, or (3) use complex sentences (clauses) to subordinate ideas. If the writer chooses to use complex sentences, they should be punctuated in the following manner:

independent clause dependent clause (no comma except when using *while* for strong contrast)

dependent clause, independent clause (comma)

Directions: With a peer, identify the different sentence types in the following text.

Telemedicine allows medical practitioners to help their patients in a number of innovative ways. A surgeon can now operate on a patient in remote locations. Telesurgery, which enables a surgeon to assist medics on a battlefield, is one example. Telemedicine is also being used for local as well as remote surgery. An example is in endoscopic surgery, during which surgeons manipulate instruments by viewing a television monitor while manipulating a tool which is inserted through a tube into a patient. Even dentists are using 3-D eyeglasses to divert a patient's attention while sitting in the chair, thus reducing anxiety or stress.



Focusing on Introductions

Nearly all introductions perform three functions:

1. interest the reader in the text,
2. give background information, and
3. present a thesis.

In order to attract the reader's attention, a writer may use dramatic description, striking words, statistics, an interesting fact, a quotation, or a question. Next, the writer often focuses on background information, which is often a reminder of information the reader already knows. The third part is the thesis statement, which contains the main idea, the writer's approach, and often a list of the main supports.

Directions: Look at the following introductions. How did the writer attract the reader's attention? Did the writer focus on background information? Is there a thesis statement?

The planet Mars has intrigued humans since we first began to study the sky. It is the only planet that looks red to the naked eye, which may explain why the ancient Greeks and Romans associated it with the bloody god of war. Spacecraft that visited Mars in the 1960s and 1970s found a freezing, arid, dead world. Nevertheless, the idea of life on Mars, and, in particular, intelligent life on Mars, fed by years of science fiction stories, persists in the public mind, despite the weight of scientific evidence against it.

Along with the many new information products flooding the marketplace, we can see a transformation in the way young people and adults come to know and understand their world. With the shift toward electronic media and information, the challenge of knowing and comprehending is complicated by a movement toward superficial presentations. Deep thinking, deep reading, and deep commentary are replaced in many situations by *sound bites*, *mind bites*, *eye candy*, and *mind candy*.

Telemedicine has been defined as the use of telecommunications to provide medical information and services. It may be as simple as two doctors discussing a case over the telephone, or as sophisticated as using satellite technology to broadcast a consultation between hospitals in two countries using videoconferencing equipment. The first is used daily by most health professionals, and the latter is used largely by medical centers.



ORAL DISCUSSION

1. Explain "immunotherapy."
2. Which predictions do you find most feasible?
3. Which predictions are partially true today?
4. Which prediction is your favorite? Why?
5. Can you add any other predictions of your own? Tell the class.

Geosciences

WARM UP

Explain the following statement: Changes in people's lifestyles necessitate changes in the workplace.

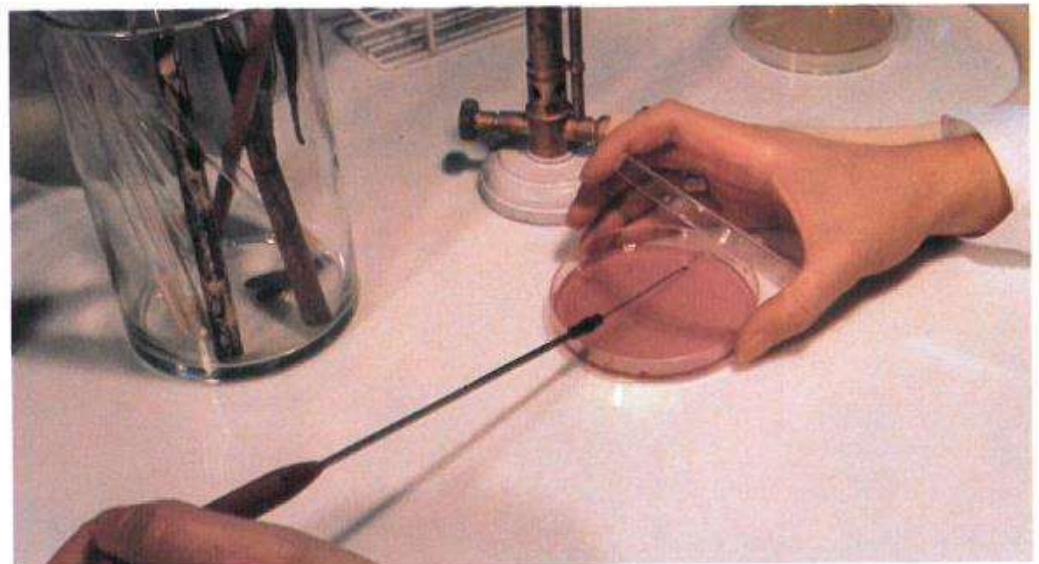
Listening Comprehension:

Listen to the following explanation of geoscientists. Prepare to be able to summarize the main idea of what you hear.

Then, answer the questions below.

Oral Discussion:

1. What earth materials would you consider the most essential to find resources for?
2. Can you think of other duties that geoscientists would be involved in?



Many nouns are formed by adding a suffix to the verb. The most common noun suffixes are *-ion* and *-ance*. Pick out the nouns from the glossary on the following page which end with these suffixes.

What is the root of each of those nouns?

What are the noun forms of *dispose*, *sustain*, and *contaminate*?

Grains: n. very small particles.
 Sustain: v. keep; maintain.
 Contaminated: adj. impure; here, poisoned.
 Fracturing: v. breaking up.

Oceanographers	investigate the physical, chemical, biological, and geologic dynamics of oceans.
Paleontologists	study fossils to understand past life forms and their changes through time and to reconstruct past environments.
Petroleum geologists	are involved in exploration for and production of oil and natural gas resources.
Petrologists	determine the origin and natural history of rocks by analyzing mineral composition and grain relationships.
Planetary geologists	study planets and their moons in order to understand the evolution of the solar system.
Sedimentologists	study the nature, origin, distribution and alteration of sediments, such as sand, silt, and mud. Oil, gas, coal, and many mineral deposits occur in such sediments.
Seismologist	study earthquakes and analyze the behavior of earthquake waves to interpret the structure of the Earth.
Soil scientists	study soils and their properties to determine how to sustain agricultural productivity and to detect contaminated soils.
Structural geologists	analyze Earth's forces by studying deformation, fracturing , and folding of the Earth's crust.
Volcanologists	investigate volcanoes and volcanic phenomena to understand these natural hazards and predict eruptions.

Oral Discussion:

Which of the above scientists would you most probably find:

- at the site of a total eclipse?
- near an active volcano?
- at a water dam?
- at a weather station?
- in an architectural firm?
- in an office in the Ministry of Petroleum in Kuwait?
- in a mine?
- on a research ocean vessel?
- in an arctic research center?
- at the Istanbul Earthquake Center in Turkey?
- at an archaeological site?
- in the middle of a desert?

Chapter 3

TECHNOLOGY

New Worlds - Part A

The Coming 500 Years

WARM UP

Discuss in class new inventions and discoveries which, say, 20 years ago were not heard of. Cellular phones, wide-screen TVs, powerful personal computers, cable TV, etc.

English hyphenates two or more words together to form new words. Mounted on the wall becomes wall-mounted. Give other similar structures that you know.

READING

What's Next?

In the box below are predictions adapted from *Time* on possible new inventions and discoveries in the near future.

READING STRATEGIES: Glossary

Wall-mounted: adj.
hung on the wall.
anti-collision: n.
Against hitting or
striking; here, car
accidents.
Detect: v. find out.
imminent: adj. about
to happen.
Permanent: adj. last-
ing for a long time.

2000	New advances in genetics enable doctors to combine gene therapy with immunotherapy to create more effective cancer treatments.
2001	Wall-mounted , one-meter long flat screens show television programs or videos, and when not in use display works of art.
2003	Mobile phones with video cameras and screens enable people to watch films or play computer games from their home or office.
2005	Video vacation postcards, postcard-sized film screens that display 10 seconds of holiday sights or sounds, are introduced onto the market.
2005	The active contact lens, linked to the Internet, allows the wearers to read e-mail without even opening their eyes.
2006	Clothes made from "smart" fabric automatically warm up people in cold weather and cool them down in hot weather.
2007	New cars are equipped with anti-collision radar and on-board computers that detect and warn drivers about imminent faults.
2010	Robotic pets, programmed to recognize their master's voice and face, operate and control all the computerized functions of the household.
2015	The genetic roots of all diseases are identified.
2017	Human beings land on Mars. A permanent colony is established on the planet around 2044.
2025	Computers connected directly to the brain are able to recognize and respond to thoughts.
2030	Doctors create fully functional artificial eyes.
2500	From an average of 75 years, human life-spans are extended to 140 years.

READING

Future Careers

Atmospheric scientists	study weather processes, the global dynamics of climate, solar radiation and its effects, and the role of atmospheric chemistry in ozone depletion , climate change, and pollution.
Economic geologists	explore for and develop metallic and nonmetallic resources. They study animal deposits and find environmentally safe ways to dispose of waste materials from mining activities.
Engineering geologists	apply geological data, techniques, and principles to the study of rock and soil materials and ground water. They investigate geologic factors that affect structures such as bridges, buildings, airports, and dams.
Environmentalists	study the interaction between the geosphere, hydrosphere, biosphere, and human activities. They work to solve problems associated with pollution, waste management, urbanization , and natural hazards, such as flooding and erosion.
Geochemists	use physical and inorganic chemistry to investigate the nature and distribution of water and to trace elements in ground water and Earth materials. They use organic chemistry to study the composition of fossil fuel (coal, oil, and gas) deposits.
Geologists	study the materials, processes, products, physical nature, and history of the Earth.
Geophysicists	apply the principles of physics to studies of the Earth's interior and investigate Earth's magnetic, electric, and gravitational fields.
Hydrogeologists	study the occurrence, movement, abundance , distribution, and quality of subsurface waters and related geologic aspects of surface waters.
Hydrologists	are concerned with water from the moment of precipitation until it evaporates into the atmosphere or is discharged into the ocean. For example, they study river systems to predict the impacts of flooding.
Marine geologists	investigate the ocean-floor and ocean-continent boundaries. They study ocean basins , continental shelves, and the coastal environments on continental borders.
Meteorologists	study the atmosphere and atmospheric phenomena, including the weather.
Mineralogists	study mineral formation, composition, and properties.

READING STRATEGIES: Glossary

Depletion: n. lessening greatly; here, of the ozone layer.

Dispose of v: get rid of.

Urbanization: n. the forming of town and cities.

Abundance: n. great quantity.

Subsurface: adj. under the surface.

Precipitation: n. here, rain.

Impacts: n. here, effects.

Basins: n. here, areas.

Expanding Your Point of View

Section 1: **Peer Feedback**

Get feedback from your peers. Get into groups of three or four and read your essay to the group. Follow the "Peer Feedback Guidelines" in your Workbook.

Section 2: **Revising**

Along with your peer feedback comments, use the following checklist to critically examine your own essay. Your teacher will also use it to evaluate your final draft.

Essay Assessment Guidelines

Content and Ideas

- _____ There is a clear thesis statement.
- _____ There is enough logical support.
- _____ There is unity.

Organization and Form

- _____ There is a clear beginning, middle and end.
- _____ It is clear which are the main supporting points and which are secondary points.

Language Conventions

- _____ There are no fragments, comma splices, run-on sentences.
- _____ There are no problems with subject-verb agreement.
- _____ There is consistency in tense, person, number, and tone.
- _____ There is a variety of sentence types.
- _____ There is coherence.
- _____ There is acceptable word choice.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: **Conferencing with the Teacher**

Section 4: **Further Drafting**



Focusing on Process/Product Writing

Section 2: Pre-writing Activities

A. Choosing a Topic

Choose a topic for the chapter writing assignment from the following list. Focus on the main organizational types which are indicated below. You will develop your topic into an essay later.

1. Definition/Reason/Result: Choose one of the careers in the geosciences that has been discussed in class. Describe it briefly. Then give reasons why one might choose this career and how you could help your own society or the world, in general, if you were to take up this career.
2. Description/Result: Review the predictions in *What's Next*. Envision another possible invention or discovery. Describe it and suggest how it might benefit your own society or the world in general.
3. Additional topic: Choose a topic that you and your teacher agree on related to technology.

B. Brainstorming



C. Paraphrasing



D. Reporting and Summarizing



E. First Draft



Writing Skills: Making Logical Assertions

When writing formally, it is best to be as objective as possible. One way is to make sure that your thesis statement makes a logical assertion.

What is a logical assertion?

A logical assertion is a statement of opinion or judgment that can be supported logically.

Directions: One of the following sentences is a logical assertion. Find it. Then, categorize the other sentences in one of the following ways. Work in groups and reach an agreement on each of the categories.

- a. a matter of judgment or opinion
- b. a matter of fact
- c. a matter of taste
- d. a possible but unverified fact.
- e. a logical assertion

1. There is intelligent life in other solar systems.
2. A volcanologist investigates volcanoes and volcanic phenomena in order to understand these natural hazards and predict eruptions.
3. I like hiking better than deep-sea diving.
4. Geoscientists use their knowledge to improve the quality of human life.
5. A woman would make a better environmental geologist than a man.

Language Conventions: Adding Coherence

In order to make your writing more logical, you must also concentrate on coherence.

What is coherence?

Coherence makes the relationships between the events and ideas clear.

The following text lacks coherence and is, therefore, hard to understand.

Focusing on Process/Product Writing

Section 1: Focusing on Conclusions

The purpose of a good conclusion is to leave the reader with a comfortable sense that the paper has been completed. There is no one "correct" way to conclude but in writing on scientific topics, the following three methods are often used singularly or in combination with one another: (1) restating/summarizing, (2) drawing additional implications, and (3) using a short, relevant quotation.

A. Restating/Summarizing

Try to restrict yourself to the most important points so that the reader can reinterpret the thesis or main idea. Many writers paraphrase their own thesis statement or main idea sentence.

B. Drawing additional implications

You may wish to suggest certain implications of your thesis or refer to some aspect of the subject which you did not refer to in your paper. For example, you may refer to ongoing research in the field, make predictions for the future, or suggest possible applications of a theory.

Do not contradict your own thesis or open up an entirely new area of the subject, which will only serve to confuse your reader.

C. Using a quotation

A relevant quotation from an expert on the topic you are discussing is always useful but best combined with other techniques. A quotation must, of course, be relevant. That is to say that it must say what you want it to say. If not, do not use it.

In a long essay, the writer often uses transitional phrases like "in conclusion," but in short paragraphs or essays, the following expressions are more commonly used.

Thus
In short,
As we have seen,
Therefore,



A Day in the Life of an Environmental Geologist

Every day is different for me in environmental consulting. Tasks change daily. Many projects are only weeks long. Now most of my time is spent in the office. A few engineers, geologists and scientists conduct my field assignments for my projects. I spend my time talking with my clients. I review reports. I keep track of money spent. I make sure that others have enough to do. Sometimes I travel to a job site. I check on the progress of the fieldwork. Most of my sites are within driving range. My long distance travel is limited to planning meetings and organizing work groups. I try to solve particular technical problems.

Coherence may be achieved by adding some of the following cohesive devices:

1. pronouns referring back to a clear antecedent	The evidence was inconclusive. It was rejected.
2. coordinating conjunctions	and, but, yet, for, so, or, nor
3. transition words and phrases	however, in addition, etc.
4. repetition of important key words	See <i>project</i> in the paragraph below.
5. subordinating conjunctions	because, although, since, etc.
6. verb tenses	The present tense is used for facts in the paragraph below.
7. verbal phrases	Being a project manager, (sentence 4 in the paragraph below)
8. synonyms	try, attempt

Directions: Read the revised text below and locate the different cohesive devices used. Work in pairs. Hint: Try to find examples of 4, 5, 7 from the list of cohesive devices suggested above.

A Day in the Life of an Environmental Geologist

Every day is different for me in environmental consulting. Tasks change daily because many projects are only weeks long. Most of my time is spent in the office, while a few engineers, geologists, and scientists conduct field assignments for my projects. Being a project manager, I spend my time talking with my clients, reviewing reports, tracking money spent, and making sure that others have enough to do. Sometimes I travel to a job site to check on the progress of the fieldwork. As most of my sites are within driving range, my long distance travel is limited to planning meetings and organizing work groups, trying to solve a particular technical problem.

A. Chapter Assessment

Your teacher will assess your essay using the guidelines found in this chapter.

B. Sequenced Research Project**Step 3: Writing the Report**

Keep your purpose in mind. You may want to simply inform your reader about group opinions on your topic. On the other hand, you may have been surprised by your findings and so you may want to share your surprise(s) with the reader. Perhaps your survey exposed a situation that you feel needs changing.

Begin your report with a statement of the general results of your survey and a description of your sample (what type of people were questioned, how many were involved, etc.). Then discuss your data in a way that seems logical to you. For example, you might organize in the following manner.

All those favoring _____
Reason 1
Reason 2

All those opposed _____
Reason 1
Reason 2
Reason 3

All those with no opinion

If you have a sampling of more than 20, you may use percentages. But if your sampling is less than 20, it would be best to use numbers like *4 out of 10 polled stated ...*

Write your first draft after you have collected your data, draw your conclusions and decide on what to include and how to organize your report.



Leaning: n. being in a position that is not upright.

Tilting: n. sloping.

Clasp: v. hold together.

leaning or **tilting** your head to one side (a problem for frequent phone users). Be aware of your posture as you sit at your computer or desk, stand in line, or carry a bag. Many of the positions we put ourselves in tighten neck and back muscles—leading to tension headaches.

Stretching and exercising therapy: Gentle neck and shoulder exercises may be used to relax and stretch strained muscles. This can reduce tension and decrease the risk of headaches triggered by muscle irritation. The following basic exercises are effective for both tension and migraine headache sufferers.

Neck Rotation

Keeping your head level, slowly turn it until you are looking straight out over one shoulder. Hold this position for 1 to 2 seconds, then gently move your head to look down at this shoulder. Return your head to center, then repeat the procedure with your other shoulder. Repeat the entire process 5 times for each shoulder.



Shoulder Retraction

With your fingertips touching behind the tops of your ears and your elbows raised, attempt to squeeze your shoulder blades together. Make sure not to push or pull on your neck. Hold the position for 5 to 10 seconds, and then release.



Upper Back Stretch

With your arms extended in front of your body, **clasp** your hands together. Gently pull your shoulder blades apart, then drop your chin to your chest. Hold this position from 10 to 30 seconds.



Neck Retraction

Squeeze your shoulder blades together. Then gently move your head straight back, making sure to keep it level. Hold this position for 5 to 10 seconds. Then relax. Repeat 10 times.



Neck Retraction

Squeeze your shoulder blades together. Then gently move your head straight back, making sure to keep it level. Hold this position for 5 to 10 seconds. Then relax. Repeat 10 times.



Chapter 1

CURRENT CONCERNS

READING STRATEGIES: Glossary

Vascular: adj. related to vessels through which liquids move in the bodies of animals or plants.

Misalignment: n. not formed or arranged in a line.

Constipation: n. the condition of not being able to empty the bowels frequently enough.

Exerts: v. uses strength or force.

Numbness: n. inability to feel anything.

Stiff: adj. painful when moving or moved.

Sore: adj. aching from infection.

The World Within Us - Part A

Headaches

WARM UP

1. What situations resulted in giving you a headache?
2. What did you do about the headaches?

READING

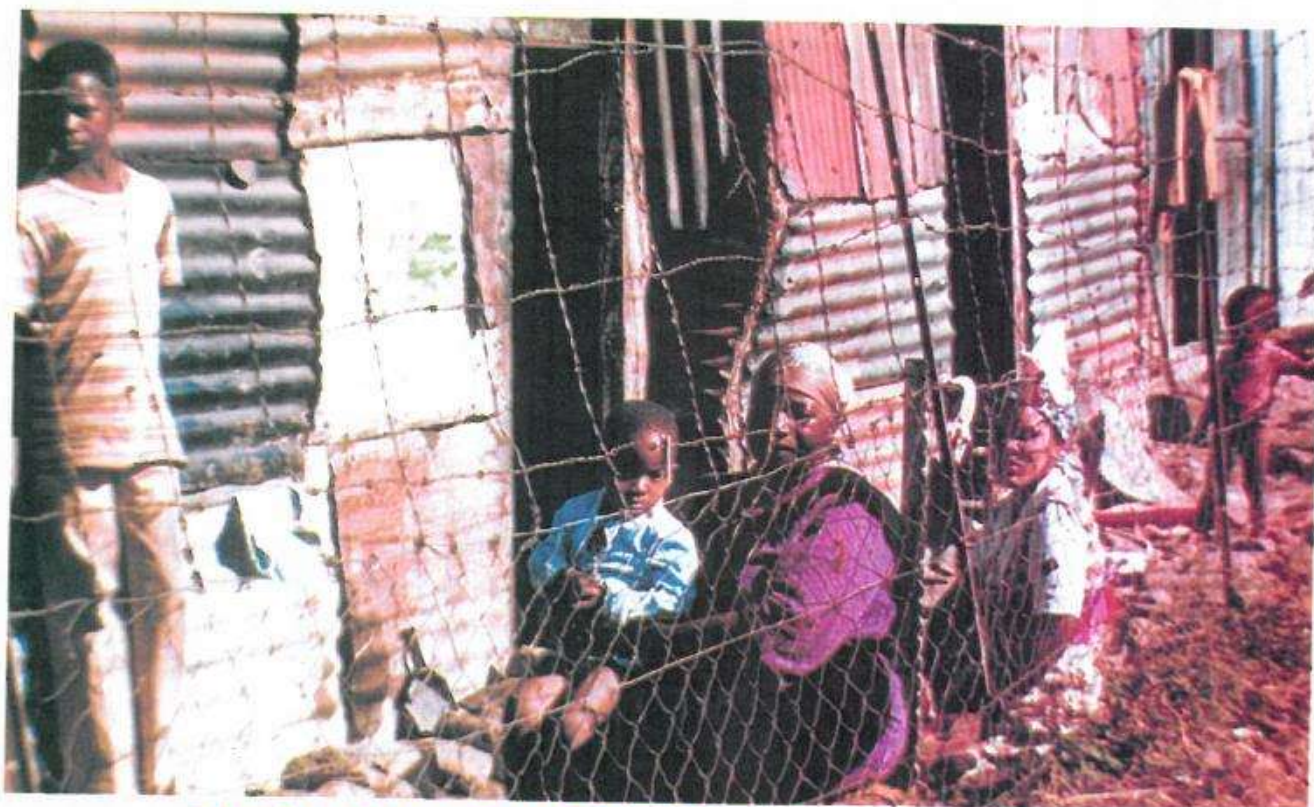
The Nature of Headaches

Headaches are caused by two different categories: muscle contractions or **vascular** irregularities. The attributes causing the headache from muscular contraction are two mechanisms. The first is nerve compression within the muscle caused by poor posture, spinal **misalignment**, or physical and emotional stress. The second mechanism is nerve irritation caused by a buildup of metabolic wastes resulting from decreased blood circulation due to poor diet, **constipation**, or other digestive problems.

On the other hand, headaches caused by vascular irregularities mainly include migraines. With vascular headaches, the alternating constriction and expansion of the arteries of the head **exerts** pressure on arterial nerves and causes sharp pain.

Symptoms to Take Seriously

Persistent headaches that progressively worsen over days or weeks
Headaches that start suddenly after coughing, straining, or exertion
Changes in vision
General weakness or **numbness**
A **stiff** neck with a fever
Unexplained breathing problems which accompany the headache
Headache after a **sore** throat or respiratory infection
A constant headache with no relief
Persistent or severe vomiting
Experiencing three or more headaches a week



Unit III: CURRENT CONCERNS

Unit Opener

"Every gun that is made, every warship launched, every rocket fired, signifies, in the final sense, a theft from those who hunger and are not fed, those who are cold and are not clothed. This world in arms is not spending money alone. It is spending the sweat of its laborers, the genius of its scientists, the hopes of its children. This is not a way of life at all in any true sense. Under the clouds of war, it is humanity hanging on a cross of iron."

Eisenhower, 1953

"Necessity is the plea for every infringement of human freedom. It is the argument of tyrants; it is the creed of slaves."

William Pitt

<i>Headache Triggers</i>	Stress
	Sudden weather changes
	Hormonal changes
	Missing a meal
	Exposure to bright light
	Too little sleep
	Exposure to cigarette smoke
	Too much sleep
	Exercise or exertion
	The changing of the seasons

Headache Treatment

Non-medicinal Therapies:

Cold therapy: An ice pack can be one of the most effective non-drug treatments. The earlier you use an ice pack to treat your headache, the better. Besides applying it to the painful area, try placing it on the back of the neck, forehead, and **temples**.

Warm therapy: Using heat to **foster** relaxation increases blood flow and relaxes your muscles. Take a warm shower, letting the water beat down on your head, neck, shoulders, and back. Hot tubs and heating pads can have a similar result. Experiment with both cold and warm treatments to see which works best for you.

Time out therapy: Remove yourself from a stressful situation by taking a walk or by taking a few minutes to concentrate on something relaxing.

Deep breathing therapy: Take deep, slow breaths, concentrating on breathing from your diaphragm. Hold your breath for several seconds, then slowly exhale, completely emptying your lungs. Repeat this several times. As you inhale, your chest and stomach areas should expand; as you exhale, these areas should contract. This increases the amount of oxygen in the body.

Proper nutrition therapy: Missing a meal and other improper eating habits can bring on a headache. If you have not had time to eat properly, get a healthy snack, such as fruit, or even a glass of water.

Botanical/Herbal medicine therapy: Many herbs claim to dull the symptoms of headaches. The English suggest belladonna, bryonia, and nux vomica as therapeutic herbs. Belladonna is commonly used to treat the violent, **throbbing** pain of migraines that are **aggravated** by light, noise, touch, a strong or unusual smell, or motion, and are characteristically worse in the afternoon. It is also commonly prescribed for headaches associated with high fever.

Good posture therapy: Posture plays a key role in many tension headaches. Avoid

Temples: n. the two sides of the forehead.
 Foster: v. help to develop.
 Throbbing: adj. beating strongly and rapidly.
 Aggravated: v. made worse.

- ✎ Begin an enjoyable task or project immediately after eating a meal.
- ✎ Call a friend who knows about your problem and have him or her just listen.
- ✎ Get enough rest and expand positive relationships.
- ✎ Carry food to school rather than buying it there. Pack healthy, satisfying food.
- ✎ Take deep breaths, close your eyes, picture yourself in a field or at the beach.
- ✎ Turn on quiet music.
- ✎ Getting on the scale does not help. It is what is inside that counts.
- ✎ Learn your danger zones.
- ✎ Drink a glass of water.

Oral Discussion

1. Cite instances where an eating disorder has hampered functioning.
2. In your words, explain the terms anorexia, bulimia, and binge eating.
3. Explain: "Disordered eating can lead to an eating disorder."
4. What is the basic difference between binge eating and bulimia?
5. Do you know any person who has any of the three types of eating disorders discussed in the text? Tell the class what effects the disorder has had on this person.
6. Which of the consequences listed in the table can be considered short term? Which can be considered long term? In pairs or groups, make another table with two columns: one for short-term consequences, the other for long term-consequences. Make your lists, then produce a common class list that can be posted on the classroom bulletin board.
7. Have you tried any of the suggestions about helping a friend or those about helping yourself? If so, tell the class about the positive results.

Chinning

Starting with your head level and looking straight ahead, inhale and **tuck in** your chin. Then exhale and stick out your chin. Repeat 3 to 5 times. This exercise helps your head stay in the proper position on your neck.



Tuck in: v. pull towards the body.

Chinning

Starting with your head level and looking straight ahead, inhale and tuck in your chin. Then exhale and stick out your chin. Repeat 3 to 5 times. This exercise helps your head stay in the proper position on your neck.



(Try to do the exercises in class.)

Headache Diary

Proper treatment of headaches involves careful diagnosis to locate the disturbance actually causing the headache. It is very important to keep track of your headaches so your doctor can properly diagnose your headache type and causes. You can use the following record:

Date:	
Warning Signs:	
Time Pain Began:	Time Pain Ended:
Type of Pain:	
Intensity of Pain:	(Low) 1 2 3 4 5 6 7 8 9 (High)
Location of Pain:	
Treatment/Medication Taken:	
Effect of Treatment:	
Hours of Sleep:	What I ate today:
Unusual Events Occurring:	

BE SMART, CHECK WITH YOUR PHYSICIAN.

Oral Discussion

1. Give examples of "poor posture" and "spinal misalignment."
2. What herbs have you heard your mother or grandmother suggest for a headache?
3. Which of the treatments listed in the lesson have you tried? Which have been effective?

For more information, try these addresses on the Internet:

www.achenet.org (American Council for Headache Education)

www.diamondheadache.com (Diamond Headache Clinic—created in 1972 to aid headache sufferers)

www.netdoor.com.entinfo/sinusaao.html (Facts about sinus pain, drainage, and infection)

www.childmedlib.org (Links to more information and organizations)

Chapter 1

The World Within Us - Part B

Eating Disorders

Warm Up

What do you consider an eating disorder to be?



READING STRATEGIES: Glossary

Crippling: adj. making a person unable to use his limbs.

Reading

Eating Disorders

Eating disorders are mental disorders that can have serious physical complications. These disorders may make normal functioning difficult and can become chronic, **crippling** illnesses, which, in extreme cases, may require hospitalization.

There are three main types of eating disorders: anorexia, bulimia, and binge eating. These are complex disorders focusing on issues of eating, body weight, and body shape. Disordered eating can lead to an eating disorder. It is important to identify eating habits and change them before they become severe.

motes poor body images which in turn may lead to disordered eating patterns and eating disorders. Other identified triggers are family dynamics and a poor self-image.

- I'm not perfect. No one will love me. Will I be good enough?
- Am I popular? Why doesn't my family notice me? I need to be the best to earn their love.
- I should be thin to be popular with my peers. Everyone diets all the time.

In our culture, physical attractiveness contributes significantly to success in the society. Thinness is an important component of how attractive and desirable a woman is perceived to be. Social pressure to be thin promotes poor body images, which in turn may prompt dieting efforts.

Treatment of eating disorders depends on a number of factors: the person with the eating disorder, the length and severity of the eating disorder, and the physical health of the person. Treatment addresses the physical and psychological aspects of the person.

Persistently: adv.
without stopping.

If you think a friend has an eating disorder, this is what you should do:

- ✎ Speak up! Approach your friend gently but **persistently**. Listen sympathetically but warn emphatically.
- ✎ Encourage your friend to talk about his or her problems and feelings.
- ✎ Be supportive. Show your friend that you believe in him or her.
- ✎ Do not focus on achievements; instead, talk about inner qualities and strengths.
- ✎ Share your own failures and mistakes.
- ✎ Do not expect that your friend will admit that he or she has a problem right away. Give it time.
- ✎ Do not discuss weight, eating habits, or exercise habits. These are results, not causes.
- ✎ Do not tell a person with anorexia or bulimia to "just go ahead and eat."
- ✎ Do not make statements about weight gained or lost. Focus on mental states.
- ✎ Be a good role model: Practice healthy living.

What to do instead of bingeing or purging?

- ✎ Go for a walk.
- ✎ Relax. Practice meditation.
- ✎ Practice new behaviors and activities, such as hobbies.
- ✎ Make new rules, such as "No eating in the car," etc.
- ✎ Do not eat while you are doing something else, such as reading, studying, watching TV, etc.

Compulsive eating or binge eating is when a person consumes a large amount of food in a short period of time (less than 2 hours), but does not engage in **purging** behavior. This eating is without regard to physical cues signaling hunger or satisfaction. The individual loses the power of choice over food.

Bulimia is when a person eats a large quantity of food in a short period of time. Then, the person attempts to get rid of the food by some method such as by vomiting, using a **laxative** or intense exercising.

Anorexia nervosa is when a person actively tries to maintain an abnormally low weight through severely **restricting** the intake of food. The person has an intense fear of gaining weight and becoming obese. A person may also engage in purging behaviors after eating even a small amount of food.

It used to be thought that eating disorders primarily occurred in middle class women. Although females are more at risk, the number of males with eating disorders is rising. Certain other groups in which eating disorders occur more frequently are dancers, gymnasts, runners, wrestlers, weight lifter, models, cheerleaders, and teenagers.

Eating disorders can have a serious and even **fatal** impact on the body. The mortality rates may not accurately reflect the number of people who died from eating disorders because of the long-term effect on the body. Even after the person no longer has an eating disorder, the damage to the body **persists**. The following table presents the effects on the body of eating disorders.

BINGE EATING	BULIMIA	ANOREXIA
Obesity	Damage to gastrointestinal system	Osteoporosis
Heart disorders	Erosion of teeth enamel	Nutritional deficiencies
Respiratory disorders	Enlarged salivary glands	Severe dehydration
Increased risk of diabetes	Chronic loss of bodily fluids leading to dehydration	Skin: dry, cracked, rough, gray, or yellow
Increased risk of high blood pressure	Lung irritation	Nails: brittle
	Stomach spasms	Hair: thin and dry
	Electrolyte imbalance that may lead to heart failure	Bruises easily
	Skin rashes	Sensitivity to cold
	Menstrual irregularities	Temporary or permanent infertility
	Death	Cardiac irregularities
		Death

For each individual, the cause of the eating disorder will be different. Several broad causes have been identified including culture and family. In our culture today, thinness is emphasized. It is **portrayed** as a key to happiness. This perception pro-

Purging: adj. getting rid of something.
Laxative: n. a medicine that causes the bowels to empty easily.

Restrict: v. limit.

Fatal: adj. deadly.

Persist: v. continue in spite of opposition or warning.

Gastrointestinal: adj. related to the stomach and intestines.

Osteoporosis: n. a disease of the bones.

Enamel: n. the hard smooth outer surface of the teeth.

Brittle: adj. hard but can be broken easily.

Spasm n. a sudden uncontrolled; tightening of the muscles.

Electrolyte: n. a liquid that can be broken down into its chemical parts by passing electricity through it, such as copper sulfate.

Rash: n. red spots on the body caused by an illness.

Portrayed: v.

described; shown.

headaches may occur every day, or almost every day, for months or years. They may last from several hours to all day. Emotional stress is the leading trigger of tension headaches, particularly among chronic sufferers. Tension headaches may also be triggered by physical causes such as poor posture or staying in one position for long periods of time. Lack of sleep is also a common contributor. This is just one aspect of the physical and emotional toll that working more and sleeping less takes on most of us. People who do not get enough sleep often find it harder to handle the stresses of everyday life. This, in turn, generates even more stress, which frequently interferes with getting to sleep or sleeping through the night. Tension headaches may be treated effectively with over-the-counter pain as well as non-medicinal therapies. If chronic tension headaches occur everyday or almost everyday, it is important that you consult a physician immediately.



Generic Nouns represent a typical member or any member of a group, such as a typical student.

Example: Every student must put in long hours of study if *he* or *she* wants to excel. The same three options for correction apply for indefinite pronouns and general nouns.

Treat collective nouns as singular unless the meaning is clearly plural. Collective nouns include words such as *jury*, *committee*, *crowd*, *family*, *audience*, *couple*, *troop*, *team*, and *class*. Ordinarily the group functions as a unit, so the noun should be considered singular. However, if the members of the group function as individuals, the noun should be treated as plural.

Examples: The committee has reached *its* decision.
The committee clapped *their* hands.

Compound antecedents connected by *and* should be treated as plural.

Example: Mariam and Samir were late handing in *their* final drafts.

When compound antecedents are connected by *or*, *nor*, *either...or* or *neither...nor*, make the pronoun agree with the nearer antecedent.

Examples: Neither the students nor their teacher could remember *his* new schedule.
Neither the teacher nor the students could remember *their* new schedule.



it probably will not.

- ___ 8. Since developed countries caused the inequality among nations, they have to solve their problems.
- ___ 9. If I make an exception for you, then I have to make an exception for everyone.



Language Conventions Pronoun agreement

Frequent misuse of plural pronouns occurs with two types of singular antecedents: indefinite pronouns and generic nouns.

Indefinite pronouns refer to nonspecific persons or things. They include:

any
either
everything
no one
each
anybody
everybody
neither
someone
anyone
everyone
none
something

Example: In class *everyone* performs at *his* or *her* level of ability.

A **pronoun** is simply a substitute for a **noun**. The pronouns or nouns that they refer to are called **antecedents**. A pronoun and its antecedent are in agreement if they are both singular or both plural. To correct a plural pronoun referring to a singular indefinite pronoun, you can do one of three things:

Replace the plural pronoun with *he* or *she* or [*his* or *her*].

Example: When someone comes to class late, *he* or *she* will probably try to make up an excuse.

Make the antecedent plural.

Example: When students come to class late, *they* will probably try to make up an excuse.

Rewrite the sentence so that no problem in agreement exists.

Example: A student who comes to class late will probably try to make up an excuse.

Writing Skills: Avoiding Logical Fallacies

A good writer should never insult the intelligence of his reader. Being honest and presenting a solid argument should be major goals.

Read the following descriptions of some possible kinds of logical fallacies.

- One. Argument from Ignorance:** When something is valid because it is not proved false or something is false because it is not proved true.
- Two. Either/or (Black/White) Fallacy:** One is forced to choose between one of two different alternatives. This is also known as "false dilemma."
- Three. Begging the Question:** Truth is assumed without supporting evidence.
- Four. False cause:** It is assumed that A caused B, just because B followed A.
- Five. Slippery Slope:** One thing leads to another in "a snowball effect."
- Six. Genetic Fallacy:** An assumption is based on what a person or thing is like from its origins.
- Seven. False Analogy:** A comparison contains more differences than similarities.
- Eight. Hasty Generalization:** Conclusions are drawn from too little evidence. This is also called "overgeneralization."
- Nine. Genuine but Insignificant Cause:** A genuine cause is insignificant when compared to other causes.

Direction: Read the following statements and match each one with the kind of logical fallacy it contains using the list above.

- ___ 1. Experts agree that there will be a computer in every household in 10 years' time.
- ___ 2. When you use a pressurized container, you are contributing to global warming.
- ___ 3. Life is like a bowl of cherries.
- ___ 4. This must be a good computer. It was made in Japan.
- ___ 5. A failing student will turn out to be just another dropout or end up in jail.
- ___ 6. It is common knowledge that eating disorders are difficult to control.
- ___ 7. Since scientists cannot prove that global warming will destroy this planet,

Focusing on Process/Product Writing

Section 1: Focusing on Unity

Every good paragraph, whether written singularly or as part of a longer piece of writing, has unity. Only one main idea is discussed. Moreover, each supporting sentence in the paragraph is directly related to the topic sentence. Unity is broken by either introducing a new idea or getting off topic. Which one of the following paragraphs lacks unity? Why?

Paragraph 1: Headache Triggers

While a biological condition may make you more susceptible to headaches, certain factors called "triggers" are what actually set them off. Therefore, an important step towards solving your headache problem is to analyze what was happening right before your head began to hurt. What were you doing? Did you skip a meal? Were you sunbathing? I sometimes spend six hours trying to get a good tan. Were you having an argument? A trigger can be almost any change inside or outside the body that can provoke a headache – something we see, feel, smell, hear, or eat.

Paragraph 2: Types of Headaches

In order to help alleviate your pain, you first need to recognize what type of headache you are having. Approximately 90% of all headaches are classified as tension-type (sometimes referred to as muscle contraction headaches). Another 8% are migraine or other types of headaches. Only about 2% are potentially more serious and may indicate an underlying illness or other medical condition.

Direction: Read the following excerpt from a text about headaches. With a classmate, decide where each new paragraph should begin. You should find four paragraphs related to symptoms, duration, causes, and treatment.

Tension Headaches

A dull, steady, non-throbbing pain, usually on both sides of the head is common. The pain can be mild to severe. Sometimes it feels as if your head is in the grip of a vise or that you are wearing a tight band across your forehead and temples. Pain may also extend into the neck and shoulder area. Occasional tension headaches can last from one to two hours, to all day, or longer. They often occur in the late afternoon or evening as a result of accumulated stress during the day. Chronic tension

Expanding Your Point of View

Section 1: Peer Feedback

Read your essay to a peer. Follow the "Peer Feedback Guidelines" in your Workbook.

Section 2: Revising

Along with your peer feedback comments, use the following checklist to critically examine your own essay. Your teacher will also use it to evaluate your final draft.

Essay Assessment Guidelines

Content and Ideas

- _____ There is a clear thesis statement.
- _____ There is enough logical support.
- _____ There is unity.

Organization and Form

- _____ There is a clear beginning, middle and end.
- _____ It is clear which are the main supporting points and which are secondary points.

Language Conventions

- _____ There are no fragments, comma splices, run-on sentences.
- _____ There are no problems with subject-verb agreement.
- _____ There is consistency in tense, person, number, and tone.
- _____ There is a variety of sentence types.
- _____ There are no problems with pronoun agreement.
- _____ There is coherence.
- _____ There is acceptable word choice.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: Conferencing with the Teacher

Section 4: Further Drafting



Focusing on Process/Product Writing

Section 2: Pre-writing Activities

A. Choosing a Topic

Choose a topic for the chapter writing assignment from the following list. Focus on the main organizational types which are indicated below. You will develop your topic into an essay later.

1. Cause/Effect/Description: Using the information in Part A of this chapter, discuss the causes (triggers) of most headaches and their effects (symptoms).
2. Comparison/ Contrast/Description: Using the table in Part B of this chapter, compare and/or contrast two of the three eating disorders mentioned (binge eating, bulimia, anorexia).
3. Classification / Description: Using multi-media sources (the Internet, articles, etc.), find information on different types of headaches (sinus, migraine, etc.). Then write an essay classifying these types of headaches.
4. Student Generated Topics: If you and your teacher agree on additional topics related to health concerns, list them and choose the one that interests you the most.

B. Brainstorming



C. Paraphrasing/Summarizing/Quoting



D. Reporting



E. First Draft



Chapter 2

CURRENT CONCERNS

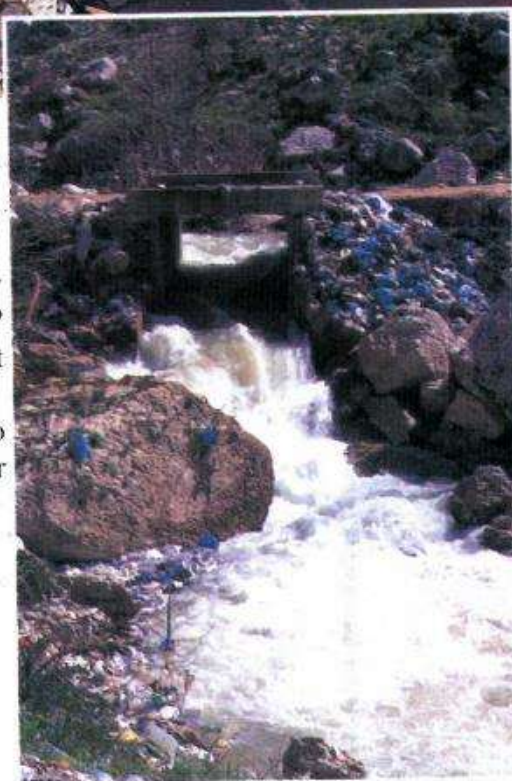
The World Around Us - Part A



Warm Up

1. What is our responsibility towards the world around us? Should we be concerned? About what?

2. What are things that we can do to make the world around us a better place?



The World Within Us - Part F

A. Chapter Assessment

Your teacher will assess your essay using the guidelines found in this chapter.

B. Sequenced Research Project

You have explored your topic in previous assignments by writing about its importance to you and to others, conducting a survey and writing a survey report. The final three steps involve interviewing an expert, summarizing documents and writing the final research report. In this chapter, you will interview an expert on your topic or someone who is very well informed about your topic.

Your Research Project

- Step 4: Interviewing an Expert
- Step 5: Summarizing Documents
- Step 6: Writing the Final Research Report

Interviewing an Expert

- Direction 1: Find an expert on your topic or someone who knows a great deal about your topic.
- Direction 2: Prepare for the interview by following the directions you followed for the survey.
Review the procedure in Unit II.
- Direction 3: Interview the expert.
- Direction 4: After the interview, write notes for yourself on where the interview took place, the nonverbal communication used by the expert, and your own impression of the interview. Review your notes or the tape. Make notes on important comments. Decide on a main idea for a report on your interview. Select support relevant to the main idea from the interview.



- Direction 5: Write your first draft. Include information explaining why the reader should consider this person an authority on your topic.



- Direction 6: Write your final draft.



Protecting the Environment

Polluted air and rivers and growing environmental hazards accompanied Japan's rapid economic growth in the 1960s. Pollution became a major social problem, causing many illnesses.



Question: What illnesses are caused by air pollution?
What illnesses are caused by river pollution?

As a result, the public began to **advocate** environmental protection.

Question: What can be done to protect the environment?

Environmental standards were established with the **enactment** of the Basic Law for Environmental Pollution Control in 1967. In 1971, the Environment Agency was set up, and efforts to prevent pollution and promote environmental protection began **in earnest**.

Home Project: Find out whether similar action was taken by the Lebanese government and when.

Companies **strove** to meet environmental standards and develop anti-pollution technologies. Gradually Japan's air and rivers became cleaner.

In the meantime, the garbage problem has become serious.

Question: Do we have a garbage problem in Lebanon? Discuss.

READING STRATEGIES: Glossary

Advocate: v. to plead in favor of; to support.
Enactment: n. legal establishment.

In earnest: in a serious and intent mental state.

Strove: v. past form of strive; to try very hard; to struggle.

Oral Discussion

1. Try to restate in your own words the reason(s) why the LEDO was established by the Lebanese government.
2. What other activities do you think the Lebanese government should adopt or initiate regarding the improvement of environmental conditions in the country?
3. What is the role of the Lebanese people in maintaining a healthy environment?

For more information about environmental issues in Lebanon:

The Center of Documentation at the Ministry of Environment in Antelias

Telephone 01-522222

Also, www.moe.gov.lb on the Internet

The Web site www.moe.gov.lb/moe/ozone will give information on the Ozone Office in Lebanon.

The Lebanese Role

Warm Up

Since Lebanon is a small country compared to the large industrialized countries, can it play a role in protecting the world?



Reading

The Lebanese Environment and Development Observatory

Although environment and development data are gradually becoming more available in Lebanon today, significant **information gaps** continue to exist in key environment—such as air and water pollution levels and industrial waste water discharges—and development areas.

Improved information on environment-development linkages is essential to **enhance** planning by government agencies, raise public environmental awareness, and increase the effectiveness of non-government organizations (NGOs), both in implementing projects at the grass-root level and in effecting policy change at the national level. Very often, environmental degradation in Lebanon can be attributed to lack of information, or of public **access** to information. In contrast, when adequate information is available and/or accessible, significant environmental improvements have been observed.

READING STRATEGIES: Glossary

Information gap: lack of certain information.
Enhance: v. increase.
Access: n. entrance.

Writing Skills: More Paraphrasing and Summarizing



Language Conventions

Making Segments Parallel

Parallel construction requires that expressions of similar content and function should be outwardly similar. The likeness of form enables the reader to recognize more readily the likeness of content and function. Parallel structure is really a matter of balance. Balancing a sentence can be compared to balancing a scale. Pretend that certain words such as **and**, **or**, and **but** are the balancing points. The words or phrases being balanced must carry the same "weight" in the sentence. One part of a sentence can be balanced only by one (or a series) of the same kind.

Examples

1. Unbalanced: I enjoy hiking in the mountains and to walk along the seafront.
Balanced: I enjoy hiking in the mountains and walking along the seafront.

To balance the sentence, the same part of speech is used for both ideas.

2. Unbalanced: Boy Scouts can learn cooking, swimming, or how to make ropes.
Balanced: Boy Scouts can learn how to cook, swim and make ropes.

3. Unbalanced: Non-traditional students often study long hours, get limited sleep, and up again early in the morning.
Balanced: Non-traditional students often study long hours, get limited sleep, and get up again early in the morning.

4. Unbalanced: The storm front was supposed to move either east towards the Bekaa or to move north towards Tripoli.
Balanced: The storm front was supposed to move either east towards the Bekaa or north towards Tripoli.

5. Unbalanced: The earthquake destroyed not only private property but also it destroyed the county's infrastructure.
Balanced: The earthquake destroyed not only private property but also the country's infrastructure.

Writers often have a mistaken belief that they should constantly vary the form of their expressions. It is true that in repeating a statement in order to emphasize it, writers may need to vary its form, but apart from this, writers should follow carefully the principle of parallel construction.



Landfill: n. a system of trash and garbage disposal in which the waste is buried between layers of earth to build up low-lying land.
Massive: adj. very big; bulky and heavy.
Contamination: n. infection; here, pollution.
Implement: v. to carry out; to accomplish.

The Ministry of Health and Welfare reports that the amount of waste households and offices produce is equivalent to each Japanese resident throwing out 1.1 kilograms of trash every day, and the mountain of garbage seems to be growing year by year. Waste produced by factories and at construction sites also continues to rise, and there is a shortage of **landfills**.

Question: Can you think of solutions for the garbage problem?

The government and civic groups are promoting recycling, and the share of newspapers and beverage cans that are recycled is reaching high levels.

Question: Where have you seen this sign?



In cooperation with other industrialized countries, Japan is also trying to address the global problems, such as pollution of seawater and the warming of the earth caused by **massive** emissions of carbon dioxide. For developing countries, it is actively providing economic and technological assistance aimed at ensuring that national development does not come at the expense of the environment.

Question: In your opinion, does Lebanon have a role to play in protecting the environment regionally? How?

What efforts are being made to make the air cleaner?

Japan enacted the Air Pollution Control Law in 1968 to halt the **contamination** of the atmosphere that was being caused by industrial development. Efforts are continuing to regulate exhaust emissions from automobiles, introduce low-pollution vehicles that run on methanol, control the amount of nitrogen oxide emitted by the diesel engines of trucks and buses, and develop devices that remove dust and sulfur from smoke emitted by factories.

In recent years, a great deal of attention has been paid to global environmental issues. Japan is actively cooperating with other members of the international community to resolve these issues. For example, when chlorofluorocarbons (CFCs), were found to be destroying the earth's ozone layer and causing global warming, Japan swiftly enacted the Ozone-Layer Protection Law in 1988. Since then, it has been **implementing** a plan to completely eliminate CFC use by the year 2000.

Research: Try to acquire a copy of the Japanese Ozone-Layer Protection Law from the Japanese Embassy. Read it in class. Can Lebanon produce something similar?

Disseminate: v. spread widely.
Stakeholders: n. here, participants.
Trends: n. general directions or courses of development.
Redundancy: n. being not needed or more than necessary.
Sustainable: adj. kept strong.

Accordingly, a coordinating unit should be established to collect, select, process, document and **disseminate** environmental data. The unit is the Lebanese Environment and Development Observatory (LEDO).

In order to maximize its benefits and added value, the LEDO must act first, as a catalyst for the collection of much-needed data. This means that the LEDO will need to work with other ministries, agencies, and academic institutions to agree on a limited number of environment and development indicators. To be successful, this process of coordination and cooperation requires that all key **stakeholders** share an equal sense of ownership of the LEDO and commitment to its successes.

The objectives of the LEDO can be summarized as follows:

1. Provide a better understanding of the situations and **trends** concerning the environment in Lebanon and key environment-development linkages, including the economic costs of environmental degradation;
2. Reduce gaps and **redundancies** in environmental data collection and analyses efforts by various public and private sector entities;
3. Supply government and non-government officials and decision makers with objective elements of information to gear their action towards **sustainable** development of Lebanon, and
4. Strengthen the capabilities of the Ministry of Environment to coordinate environmental activities by various public and private sector entities and to formulate realistic environmental policies.

In order to meet its objectives, the LEDO must fulfil the following essential conditions:

1. Have LEDO partners share a strong and genuine sense of participation in and ownership of the LEDO.
2. Work closely and coordinate with the Central Administration of Statistics, especially on environment-development linkages.
3. Build on and complement ongoing or previous institutional development activities.
4. Focus on a limited number of environment and development indicators.
5. Draw on the technical capabilities and expertise of regional agencies and programs, such as the Blue Plan and the Mediterranean Environment and Development Observatory.
6. Provide open and free access to data via the Internet.



Focusing on Process/Product Writing**Section 2: Pre-writing Activities****A. Choosing a Topic**

Choose a topic for the chapter writing assignment from the following list. Focus on the main organizational types which are indicated below. You will develop your topic into an essay later.

1. Description/Reason: Discuss one environmental problem which the Lebanese face. Describe it. Give reasons for its existence and suggest ways to control or eradicate this problem.
2. Comparison/ Description: Compare ways in which the Japanese and the Lebanese are trying to maintain a healthy environment. (garbage, clean air, etc.)
3. Process/Description: What role can a Lebanese citizen play in trying to maintain a healthy environment? Use suggestions from your class discussion in Part B of this chapter.
4. Student Generated Topics: If you and your teacher agree on additional topics related to environmental concerns, list them and choose the one that interests you the most.

B. Brainstorming**C. Paraphrasing/Summarizing/ Quoting****D. Reporting****E. First Draft**

A. Chapter Assessment

Your teacher will assess your essay using the guidelines found in this chapter.

B. Sequenced Research Project - Summarizing Documents**Summarizing Documents**

- Direction 1: Find three documents (publications) that you believe will be useful in writing your final research report. They may come from any type of multi-media material (magazines, newspapers, the Internet, etc.)
- Direction 2: Write an introductory and a concluding paragraph which link these three publications together. Link the most important or interesting information together. Remember to include a thesis statement in your introduction which makes it clear what the point of your paper is.
- Direction 3: Decide which one of your publications you will summarize first, second and third (three body paragraphs). Provide a logical link between the summaries. For example, the next summary approaches the topic from a different perspective.
- Direction 4: Ask a peer to critique your draft using the "Peer Feedback Guidelines" in the Workbook.



- Direction 5: Write the final draft.



Expanding Your Point of View

Section 1: Peer Feedback

Read your essay to a peer. Follow the "Peer Feedback Guidelines" in your Workbook.

Section 2: Revising

Along with your peer feedback comments, use the following checklist to critically examine your own essay. Your teacher will also use it to evaluate your final draft.

Essay Assessment Guidelines

Content and Ideas

- ☐ There is a clear thesis statement.
- ☐ There is enough logical support.
- ☐ There is unity.

Organization and Form

- ☐ There is a clear beginning, middle and end.
- ☐ It is clear which are the main supporting points and which are secondary points.

Language Conventions

- ☐ There are no fragments, comma splices, run-on sentences.
- ☐ There are no problems with subject-verb agreement.
- ☐ There is consistency in tense, person, number, and tone.
- ☐ There is a variety of sentence types.
- ☐ There are no problems with pronoun agreement.
- ☐ There is coherence.
- ☐ There is clear reference.
- ☐ There is acceptable word choice.
- ☐ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: Conferencing with the Teacher

Section 4: Further Drafting



in the APA but all the other lines are not.

Book

Brecht, R.C. (1999). The state of art in telemedicine. Washington, D.C.: IOS Press.

Article

Hendrix, C. E. (1995, August). Factors affecting performance in virtual environments. Popular Science, pp. 20 – 29.

When you compile a list of references, you should check a standard guidebook for other types of sources, such as newspaper articles, government documents, the Internet, and so forth. Your teacher will suggest a current guidebook for you to refer to when writing your final report for your research project.

Punctuating Quotations

Direct Quotations

Words, phrases, or two or three short sentences are quoted directly from your source and are enclosed in quotation marks. Remember that a quotation that ends with a period or a comma is always followed by quotation marks.

When you want to quote three or more long sentences, the entire quotation is indented on both the right and the left margins. No quotation marks are used, and spacing should be single.

If you choose to leave out part of the quotation, even a single word or letter, use an ellipsis (...).

Indirect Quotations

Do not use quotation marks for an indirect quotation. Simply incorporate it in paraphrased form into your text.

Writers who do not give credit to their sources are guilty of **plagiarism – that is, the stealing of ideas, words, or research and presenting them as their own.**

Focusing on Process/Product Writing

Section I: Refocusing on Citing Sources, Paraphrasing, Summarizing, Quoting

When you use another writer's idea(s) either by paraphrasing, summarizing or quoting, you must cite your source. One more formal way of citing sources in the applied sciences and the social sciences is to use the APA format.

Direct Quotations

When you use a direct quotation, the source is ordinarily cited by giving (1) the author's family name, (2) the year of publication, and (3) the page number(s). The citation is placed in parenthesis after the quotation.

Virtual reality systems "require the user to be isolated from the real world and surrounded by the virtual world generated by the computer."
(Hendrix, 1995, 21)

If, however, you have used the author's name already, only the date and page(s) are given.

Hendrix feels that "the human-computer interaction must provide a realistic environment with which the user can interact." (1995, 23)

Also note that when a direct quotation by a certain author comes soon after the author's name has been mentioned in a previous reference (that is, no other author has been mentioned in between), only the page number(s) is given.

Indirect Quotations

When you use indirect quotations (paraphrasing) or when a source is simply referred to, cite the source giving only (1) the author's family name and (2) the year of publication.

Brecht describes how virtual environments are being used in health care.
(Brecht, 1999)

Whether citing direct or indirect quotations, a writer must include a list of references at the end of his text so that the reader can find additional information if he/she so desires.

Listing References

Your list of references follows a particular format and is arranged in alphabetical order according to the authors' family names. Note that the first line is indented

Listlessly: adv. weakly, lacking liveliness.

"I felt fine!" said Em fiercely.

Lou shrugged. "Well, you've got to realize, the world wouldn't be able to support twelve billion people if it wasn't for processed seaweed and sawdust. I mean, it's a wonderful thing, really, I guess. That's what they say."

"They say the first thing that pops into their heads," said Em. She closed her eyes. "Golly—remember shopping, Lou? Remember how the stores used to fight to get our folks to buy something? You didn't have to wait for somebody to die to get a bed or chairs or a stove or anything like that. Just went in—bing!—and bought whatever you wanted. Gee whiz, that was nice, before they used up all the raw materials. I was just a little kid then, but I can remember so plain."

Depressed, Lou walked **listlessly** to the balcony's edge, and looked up at the clean, cold, bright stars against the black velvet of infinity. "Remember when we used bugs on science fiction, Em? Flight seventeen, leaving for Mars, launching ramp twelve. 'Board! All non-technical personnel kindly remain in bunkers. Ten seconds...nine...eight...seven...six...five...four...three...two...one! *Main Stage! Barrrrroooooom!*'"

"Why worry about what was going on on earth?" said Em, looking up at the stars with him. "In another few years, we'd all be shooting through space to start life over again on a new planet."

Lou sighed. "Only it turns out you need something about twice the size of the Empire State Building to get one lousy colonist to Mars. And for another couple of trillion bucks he could take his wife and dog. That's the way to lick overpopulation—*emigrate!*"

"Lou—?"

"Hmmm?"

"When's the Five-Hundred-Mile Speedway Race?"

"Uh—Memorial Day, May thirtieth."

She bit her lip. "Was that awful of me to ask?"

"Not very, I guess. Everybody in the apartment looked it up to make sure."

"I don't want to be awful," said Em, "but you've just got to talk over these things now and then, and get them out of your system."

"Sure you do. Feel better?"

"Yes—and I'm not going to lose my temper anymore, and I'm going to be just as nice as I know how."

Chapter 3

CURRENT CONCERNS

New Worlds—Part A

Warm Up

What changes (in lifestyle) do you think will result from technological advances?



Reading

Read the first part of the short story, stopping to answer or discuss the questions in the boxes.

"Tomorrow and Tomorrow and Tomorrow"

By Kurt Vonnegut, Jr.

The year was 2158 A.D., and Lou and Emerald Schwartz were whispering on the balcony outside Lou's family's apartment on the seventy-sixth floor of Building 257 in Alden Village, a New York housing development that covered what had once been known as Southern Connecticut. When Lou and Emerald had married, Em's parents had tearfully described the marriage as being between May and December; but now, with Lou one hundred and twelve and Em ninety-three, Em's parents had to admit that the match had worked well.

"That's just the way she picked to go," said Em.

"Then what was she doing carrying a sex-pack of anti-gerasone when she got it?"

Emerald shook her head wearily and covered her eyes. "I dunno, I dunno, I dunno. All I know is, something" has just got to be done." She sighed. "Sometimes I wish they'd left a couple of diseases kicking around somewhere, so I could get one and go to bed for a little while. Too many people!" she cried, and her words **cackled** and **gabbled** and died in a thousand asphalt-paved, skyscraper-walled courtyards.

Lou laid his hand on her shoulder tenderly. "Aw, hon, I hate to see you **down in the dumps** like this."

"If we just had a car, like the folks used to in the old days," said Em, "we could go for a drive, and get away from people for a little while. Gee—if those weren't the days!"

"Yeah," said Lou, "before they'd used up all the metal."

"We'd hop in, and Pop'd drive up to a filling station and say, '**Fillerup!**'"

"That was the nuts, wasn't it—before they'd used up all the gasoline."

"And we'd go for a carefree ride in the country."

"Yeah—all seems like a fairyland now, doesn't it, Em? Hard to believe there really used to be all that space between cities."

Question: Do you think that what Lou and Em are talking about is feasible? Do you see any trends that signal such a situation becoming a reality in the future?

"And when we got hungry," said Em, "we'd find ourselves a restaurant, and walk in, big as you please, and say, 'I'll have a steak and French-fries, I believe,' or 'How are the pork chops today?' " She licked her lips, and her eyes **glistened**.

"Yeah man!" growled Lou. "How'd you like a hamburger **with the works**, Em?"

"Mmmmmmmmm."

"If anybody'd offered us processed seaweed in those days, we would have spit right in his eye, huh, Em?"

"Or processed sawdust," said Em.

Doggedly, Lou tried to find the cheery side of the situation.

"Well, anyway, they've got the stuff so it tastes a lot less like seaweed and sawdust than it did at first; and they say it's actually better for us than what we used to eat."

Cackled: v. gave an unpleasant sound.
Gabbled: v. could not be heard clearly.
Down in the dumps: idiom: depressed.
Fillerup, slang: Fill her up, or fill the car with gasoline.
Glistened: v. shone; here, maybe with tears.
With the works: idiom: here, with all types of vegetables and sauces.
Doggedly: adv. with determination; without giving up.

Gloomily: adv. sadly.
Dandelion: n. type of
small wild bright-yel-
low flower; often con-
sidered a useless
plant.
Smeared; v. here:
killed, run down.

and before that the Presidential Elections, and before that I-don't-know-what. It's been just one excuse after another for fifty years now. I don't think we're ever going to get a room to ourselves or an egg or anything."

"All right—call me a failure!" said Lou. "What can I do? I work hard and make good money, but the whole thing, practically, is taxed away for defense and old age pensions. And if it wasn't taxed away, where do you think we'd find a vacant room to rent? Iowa, maybe? Well, who wants to live on the outskirts of Chicago?"

Question: What problem in the future is the author anticipating?

Em put her arms around his neck. "Lou, hon, I'm not calling you a failure. The Lord knows you're not. You just haven't had a chance to be anything or have anything because Gramps and the rest of his generation won't leave and let somebody else take over."

"Yeah, yeah," said Lou **gloomily**. "You can't exactly blame 'em, though, can you? I mean, I wonder how quick we'll knock off the anti-gerasone when we get to Gramp's age."

"Sometimes I wish there wasn't any such thing as anti-gerasone!" said Emerald passionately. "Or I wish it was made out of something real expensive and hard-to-get instead of mud and **dandelions**. Sometimes I wish folks just up and died regular as clockwork, without anything to say about it, instead of deciding themselves how long they're going to stay around. There ought to be a law against selling the stuff to anybody over one hundred and fifty."

"Fat chance of that," said Lou, "with all the money and votes the old people've got." He looked at her closely. "You ready to up and die, Em?"

"Well, for heaven's sakes, what a thing to say to your wife. Hon! I'm not even one hundred yet." She ran her hands lightly over her firm, youthful figure, as though for confirmation. "The best years of my life are still ahead of me. But you can bet that when one hundred and fifty rolls around, old Em's going to pour her anti-gerasone down the sink, and quit taking up room, and she'll do it smiling."

"Sure, sure," said Lou, "you bet. That's what they all say. How many you heard of doing it?"

Question: If you got to one hundred and five, would you decide that it was about time you died?

"There was that man in Delaware."

"Aren't you getting kind of tired talking about him, Em? That was five months ago."

"All right, then—Gramma Winkler, right here in the same building."

"She got **smeared** by a subway."

Question: What things in the first paragraph are different from our world today?

Question: Explain Em's parents' figure of speech describing the marriage between Lou and Em.

But Em and Lou weren't without their troubles, and they were out in the **nippy** air of the balcony because of them.

"Sometimes I get so mad, I feel like just up and diluting his anti-gerasone," said Em.

"That'd be against Nature, Em," said Lou, "it'd be murder. Besides, if he caught us **tinkering** with his anti-gerasone, not only would he disinherit us, he'd bust my neck. Just because he's one hundred and seventy-two doesn't mean Gramps isn't strong as a bull."

"Against Nature," said Em. "Who knows what Nature's like anymore? Ohhhhh—I don't guess I could ever bring myself to dilute his anti-gerasone or anything like that, but, **gosh**, Lou, a body can't help thinking Gramps is never going to leave if somebody doesn't help him along a little. **Golly**—we're so crowded a person can hardly turn around, and Verna's dying for a baby, and Melissa's gone thirty years without one." She stamped her feet. "I get so sick of seeing his wrinkled old face, watching him take the only private room and the best chair and the best food, and getting to pick out what to watch on TV, and running everybody's life by changing his will all the time."

"Well, after all," said Lou **bleakly**, "Gramps is head of the family. And he can't help being wrinkled like he is. He was seventy before anti-gerasone was invented. He's going to leave, Em. Just give him time. It's his business. I know he's tough to live with, but be patient. It wouldn't do to do anything that's rile him. After all, we've got it better'n anybody else, there on the daybed."

Question: What do you think "anti-gerasone" is?

"How much longer do you think we'll get to sleep on the daybed before he picks another pet? The world's record's two months, isn't it?"

"Mom and Pop had it that long once, I guess."

"When is he going to leave, Lou?"



"Well, he's talking about giving up anti-gerasone right after the five-hundred-mile Speedway Race."

"Yes—and before that it was the Olympics, and before that the World's Series,

READING STRATEGIES: Glossary

Nippy: adj. cold, chilly.

Tinkering: v. playing around.

Gosh, golly: interj. Expressions, usually of surprise.

Bleakly: adv. cheerlessly.

"That's my Em."

They squared their shoulders, smiled bravely, and went back inside.

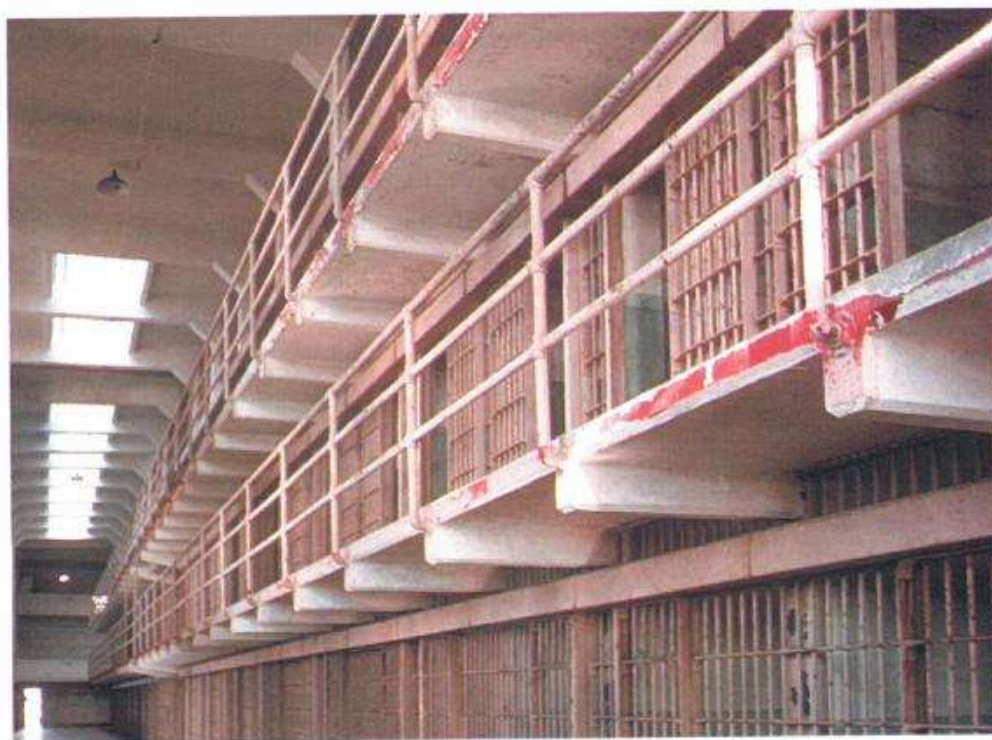
Question: What do you anticipate Lou and Em will do next?

Oral Discussion:

1. The author describes what he believes life will be like in the future. Do you agree?
2. What problems other than overpopulation and the depletion of raw materials do you think future generations may have?
3. Do you think that man will be able to prolong life? What are the advantages and disadvantages of such a medical advancement?

Warm Up

Discuss whether the happenings in the first part of the story are too fantastic or not. Give reasons and try to be as scientific as possible.



**READING
STRATEGIES:**
Glossary

Irascibly: adv. with a tendency of being angry.

Jab: v. strike quick blows.

Reading

Following is the second part of the story. As you read, try to think of answers to the questions in the boxes. Your teacher will allow you time to discuss these questions with a partner.

"Tomorrow and Tomorrow and Tomorrow" (Cont'd)

Gramps Schwarz, his chin resting on his hands, his hands on the crook of his cane, was staring **irascibly** at the five-foot television screen that dominated the room. On the screen, a news commentator was summarizing the day's happenings. Every thirty seconds or so, Gramps would **jab** the floor with his cane-tip and shout, "Hell! We did that a hundred years ago!"

Emerald and Lou, coming in from the balcony, were obliged to take seats in the back row, behind Lou's father and mother, brother and sister-in-law, son and daugh-

ter-in-law, grandson and wife, granddaughter and husband, great-grandson and wife, nephew and wife, grandnephew and wife, great-grandniece and husband, great-grandnephew and wife, and, of course, Gramps, who was in front of everybody. All, save Gramps, who was somewhat **withered** and bent, seemed to be pre-anti-gerasone standards, to be about the same age—to be somewhere in their late twenties or early thirties.

Question: Do you find the above situation strange: that everyone will someday look the same age because of a popular wonder drug?

*"Meanwhile," the commentator was saying, "Council Bluffs, Iowa, was still threatened by stark tragedy. But two hundred weary rescue workers have refused to give up hope, and continue to dig in an effort to save Elbert Haggdorn, one hundred and eighty-three, who has been **wedged** for two days in a ..."*

Question: Why is the above in italics?

"I wish he'd get something more cheerful," Emerald whispered to Lou.

"Silence!" cried Gramps. "Next one shoots off his big bazoo while the TV's on is gonna find hisself cut off without a dollar"—and here his voice suddenly softened and sweetened—"when they wave that checkered flag at the Indianapolis Speedway, and old Gramps gets ready for the Big Trip Up Yonder." He sniffed sentimentally, while his heirs concentrated desperately on not making the slightest sound. For them, the **poignancy** of the prospective Big Trip had been dulled somewhat by its having been mentioned by Gramps about once a day for fifty years.

"Dr. Brainard Keyes Bullard," said the commentator, "President of Wyandotte College, said in an address tonight that most of the world's ills can be traced to the fact that Man's knowledge of himself has not kept pace with his knowledge of the physical world."

Question: Can you explain what the commentator is saying?

"Hell!" said Gramps. "We said that a hundred years ago!"

*"In Chicago tonight," said the commentator, "a special celebration is taking place in the Chicago Lying-in Hospital. The guest of honor is Lowell W. Hitz, age zero. Hitz, born this morning, is the twenty-five-millionth child to be born in the hospital." The commentator faded, and was replaced on the screen by young Hitz, who **squalled** furiously.*

"Hell," whispered Lou to Emerald, "we said that a hundred years ago."

"I heard that!" shouted Gramps. He snapped off the television set, and his **petrified** descendants stared silently at the screen. "You, there, boy—"

"I didn't mean anything by it, sir," said Lou.

"Get me my will. You know where it is. You kids *all* know where it is. Fetch, boy!"

Withered: adj. reduced in size.

Wedge: v. caught in a place without being able to move.

Poignancy: n. sharp feeling of sadness and pity.

Squalled: v. cried unpleasantly.

Petrified: adj. scared.

heroically brought his powerful emotions under control, and, after erasing for a full minute, he began to write and speak at the same time. Lou could have spoken his lines for him, he'd heard them so often.

"I have had many heartbreaks **ere** leaving this vale of tears for a better land," Gramps said and wrote. "But the deepest hurt of all has been dealt me by—" He looked around the group, trying to remember who the malefactor was.

Everyone looked helpfully at Lou, who held up his hand resignedly.

Gramps nodded, remembering, and completed the sentence: "my great-grandson, Louis J. Schwarz."

"Grandson, sir," said Lou.

"Don't **quibble**. You're in deep enough now, young man," said Gramps, but he changed the trifle. And from there he went without a misstep through the phrasing of the disinheritance, causes for which were disrespectfulness and quibbling.

In the paragraph following, the paragraph that had belonged to everyone in the room at one time or another, Lou's name was scratched out and Willy's substituted as heir to the apartment and, the biggest plum of all, the double bed in the private bedroom. "So!" said Gramps **beaming**. He erased the date at the foot of the will, and substituted a new one, including the time of day. "Well—time to watch the McGarvey Family." The McGarvey Family was a television serial that Gramps had been following since he was sixty, or for one hundred and twelve years. "I can't wait to see what's going to happen next," he said.

Lou detached himself from the group and lay down on his bed of pain by the bathroom door. He wished Em would join him, and he wondered where she was.

He dozed for a few moments, until he was disturbed by someone's stepping over him to get into the bathroom. A moment later, he heard a faint **gurgling** sound, as though something were being poured down the washbasin drain. Suddenly, it entered his mind that Em had **cracked up**, and that she was in there doing something drastic about Gramps.

"Em--!" he whispered through the panel. There was no reply, and Lou pressed against the door. The worn lock, whose bolt barely engaged its socket, held for a second, then let the door swing inward.

"Morty!" gasped Lou.

Lou's great-grandnephew, Mortimer, who had just married and brought his wife home to the Schwartz menage, looked at Lou with **consternation** and surprise. Morty kicked the door shut, but not before Lou had glimpsed what was in his hand—Gramp's enormous economy-size bottle of anti-gerasone, which had been half-emptied, and which Morty was refilling to the top with tap water.

A moment later, Morty came out, glared defiantly at Lou, and brushed past him wordlessly to rejoin his pretty bride.

Ere, archaic: before.
Quibble: v. argue about small points.
Beaming: v. smiling brightly and happily.
Gurgling: adj. having the sound of water flowing, as out of an opening.
Cracked up: v. became crazy.
Consternation: n. shock and fear.

Dispersed: v. put in different places.

Impeccable: adj. faultless.

Portentous adj. foretelling of evil happenings.

Lou nodded dully, and found himself going down the hall, picking his way over bedding to Gramps's room, the only private room in the Schwartz apartment. The other rooms were the bathroom, the living room, and the wide, windowless hallway, which was originally intended to serve as a dining area, and which had a kitchenette in one end. Six mattresses and four sleeping bags were **dispersed** in the hallway and the living room, and the daybed, in the living room, accommodated the eleventh couple, the favorites of the moment.

On Gramp's bureau was his will, smeared, dog-eared, perforated, and blotched with hundreds of additions, deletions, accusations, conditions, warnings, advice, and homely philosophy. The document was, Lou reflected, a fifty-year diary, all jammed onto two sheets—a garbled, illegible log of day after day strife. This day, Lou would be disinherited for the eleventh time, and it would take him perhaps six months of **impeccable** behavior to regain the promise of a share in the estate.

"Boy!" called Gramps.

"Coming, sir!" Lou hurried back into the living room, and handed Gramps the will.

"Pen!" said Gramps.

He was instantly offered eleven pens, one from each couple.

"Not *that* leaky thing," he said, brushing Lou's pen aside. "Ah, there's a nice one. Good boy, Willy." He accepted Willy's pen. That was the tip they'd all been waiting for. Willy, then, Lou's father, was the new favorite.

Willy, who looked almost as young as Lou, though one hundred and forty-two, did a poor job of concealing his pleasure. He glanced shyly at the daybed, which would become his, and from which Lou and Emerald would have to move back into the hall, back to the worst spot of all by the bathroom door.

Gramps missed none of the high drama he'd authored and he gave his own familiar role everything he had. Frowning and running his finger along each line, as though he were seeing the will for the first time, he read aloud in a deep, **portentous** monotone, like a bass tone on a cathedral organ"

"I, Harold D. Schwartz, residing in Building 257 of Alden Village, New York City, do hereby make, publish and declare this to be my last Will and Testament, hereby revoking any and all former wills and codicils by me at any time heretofore made." He blew his nose importantly, and went on, not missing a word, and repeating many for emphasis—repeating in particular his ever-more-elaborate specifications for a funeral.

Question: Why do you think the grandfather is making a big thing out of reading the will?

At the end of these specifications, Gramps was so choked with emotion that Lou thought he might forget why he'd gotten out the will in the first place. But Gramps

Warm Up

By now, you should have been able to understand the main flow of events in the story. However, the climax in the plot has not been reached yet. Can you tell what it might be?

Reading

This is the third and last part of the story. Again, as in the first two parts, try to answer the questions in the boxes or discuss them with a partner, whichever your teacher instructs you to do.

"Tomorrow and Tomorrow and Tomorrow" (Cont'd)

The Schwatzes contemplated Lou in **incredulous** silence for a moment longer, and then hurried back to the living room, as though some of his horrible guilt would taint them, too, if they looked too long. Morty stayed behind long enough to give Lou a quizzical, annoyed glance. Then he, too, went into the living room, leaving only Emerald standing in the doorway.

Question: What was the meaning of Monty's "quizzical, annoyed glance?"

Tears streamed over her cheeks. "Oh, you poor lamb—please don't look so awful. It was my fault. I put you up to this."

"No," said Lou, finding his voice, "really you didn't. Honest, Em, I was just—"

"You don't have to explain anything to me, hon. I'm on your side no matter what." She kissed him on his cheek, and whispered in his ear. "It wouldn't have been murder, hon. It wouldn't have killed him. It wasn't such a terrible thing to do. It just would have fixed him up so he'd be able to go any time God decided He wanted him."

"What's gonna happen next, Em?" said Lou hollowly. "What's he gonna do?"

Lou and Emerald stayed fearfully awake almost all night, waiting to see what Gramps was going to do. But not a sound came from the sacred bedroom. At two hours before dawn, the pair dropped off to sleep.

At six o'clock they arose again, for it was time for their generation to eat breakfast in the kitchenette. No one spoke to them. They had twenty minutes in which to eat, but their reflexes were so dulled by the bad night that they had hardly swallowed two mouthfuls of egg-type processed seaweed before it was time to surrender their places to their son's generation.

reading
strategies:
vocabulary

incredulous: adj.
showing disbelief.

"Or the World's Series," said Eddie.

"Or whether Mrs. McGarvey got her eyesight back," said Morty.

Question: Do you think the family was sincerely sad at Gramps's disappearance?

"There's more," said Lou, and he began reading aloud again: " 'I, Harold D. Schwartz...do hereby make publish and declare this to be my last Will and Testament, hereby revoking any and all former wills and codicils by me at any time heretofore made.' "

"No!" cried Willy. "Not another one!"

" 'I do stipulate,' " read Lou, " 'that all of my property, of whatsoever kind and nature, not be divided, but do devise and bequeath it to be held in common by my **issue**, without regard for generation, equally, share and share alike.' "

"Issue?" said Emerald.

Lou included the multitude in a sweep of his hand. "It means we all own the whole damn shootin' match."

All eyes turned instantly to the bed.

"Share and share alike?" said Morty.

"Actually," said Willy, who was the oldest person present, "it's just like the old system, where the oldest people head up things with their headquarters in here, and—"

"I like *that*!" said Em. "Lou owns as much of it as you do, and I say it ought to be for the oldest one who's still working. You can **snooze** around here all day, waiting for your pension check, and poor Lou stumbles in here after work, all tuckered out, and—"

"How about letting somebody who's never had any privacy get a little crack at it?" said Eddie hotly. "Hell, you old people had plenty of privacy back when you were kids. I was born and raised in the middle of the goddam barracks in the hall! How about—"

"Yeah?" said Morty. "Sure, you've all had it pretty tough, and my heart bleeds for you. But try honeymooning in the hall for a real kick."

Question: Who do you think should get the bedroom: the oldest, the oldest working, the one who has never had privacy, or the honeymooners? Explain.

"Silence!" shouted Willy **imperiously**. "The next person who opens his mouth spends the next six months by the bathroom. Now clear out of my room. I want to think."

Issue: n. here: all children and grandchildren.

Snooze: v. nap, short sleep.

Imperiously: adv. with a commanding tone.

Mousetrapped: adj.
here, of bad quality.
Harrowing: adj. totally
unbearable.
Stealthily: adv. without
making a sound, like a
thief.
Imperceptible: adj.
unable to be noticed.
Dumbfounded: adj.
unable to say some-
thing because of shock
or surprise.

Shocked, Lou didn't know what on earth to do. He couldn't let Gramps take the **mousetrapped** anti-gerasone; but if he warned Gramps about it, Gramps would certainly make life in the apartment, which was merely insufferable now, **harrowing**.

Question: If you were Lou, what would you do?

Lou glanced into the living room, and saw that the Schwartzes, Emerald among them, were momentarily at rest, relishing the botches that McGarvey's had made of *their* lives. **Stealthily**, he went into the bathroom, locked the door as well as he could, and began to pour the contents of Gramps's bottle down the drain. He was going to refill it with full-strength anti-gerasone from the twenty-two smaller bottles on the shelf. The bottle contained a half-gallon, and its neck was small, so it seemed to Lou that the emptying would take forever. And the almost **imperceptible** smell of anti-gerasone, like Worcestershire sauce, now seemed to Lou, in his nervousness, to be pouring out into the rest of the apartment through the keyhole and under the door.

"Gloog-gloog-gloog-gloog-" went the bottle monotonously. Suddenly, up came the sound of music from the living room, and there were murmurs and the scraping of chair legs on the floor. "This ends," said the television announcer, *"the 29,121st chapter in the life of your neighbors and mine, the McGarveys."* Footsteps were coming down the hall. There was a knock on the bathroom door.

"Just a second," called Lou cheerily. Desperately, he shook the big bottle, trying to speed up the flow. His palms slipped on the wet glass, and the heavy bottle smashed to splinters on the tile floor.

The door sprang open, and Gramps, **dumbfounded**, stared at the mess.

Lou grinned engagingly through his nausea, and, for want of anything remotely resembling a thought, he waited for Gramps to speak.

Question: In your own words, paraphrase the paragraph above. (show Lou's reaction at being caught, why he felt nauseous, and whether you would react in the same way.)

"Well, boy," said Gramps at last, "looks like you've got a little tidying up to do."

And that was all he said. He turned around, elbowed his way through the crowd, and locked himself in his bedroom.

Oral Discussion

In groups, discuss Lou's reaction to his cousin's act. Try to come up with an alternative way of correcting the act.

Hauled: v. carried.

Fray: n. fight, quarrel.

Pipe down: v. slang:

be quiet.

Toss the whole kit and

caboodle of you right

out: slang: throw all of

you out.

A vase shattered against the wall, inches above his head. In the next moment, a free-for-all was underway, with each couple battling to eject every other couple from the room. Fighting coalitions formed and dissolved with the lightning changes of the tactical situation. Em and Lou were thrown into the hall, where they organized others in the same situation, and stormed back into the room.

After two hours of struggle, with nothing like a decision in sight, the cops broke in.

For the next half-hour, patrol wagons and ambulances **hauled** away the Schwartzes, and then the apartment was still and spacious.

An hour later, films of the last stages of the riot were being televised to 500,000,000 delighted viewers on the Eastern Seaboard.

In the stillness of the three-room Schwartz apartment on the 76th floor of Building 257, the television set had been left on. Once more the air was filled with the cries and grunts and crashes of the **fray**, coming harmlessly now from the loud-speaker.

The battle also appeared on the screen of the television set in the police station, where the Schwartzes and their captors watched with professional interest.

Em and Lou were in adjacent four-by-eight cells, and were stretched out peacefully on their cots.

"Em—" called Lou through the partition, "you got a washbasin all your own too?"

"Sure. Washbasin, bed, light—the works. Ha! And we thought Gramps's room was something. How long's this been going on?" She held out her hand. "For the last time in forty years, hon, I haven't got the shakes."

"Cross your fingers," said Lou, "the lawyer's going to try to get us a year."

"Gee," said Em dreamily, "I wonder what kind of wires you'd have to pull to get solitary?"

"All right, **pipe down**," said the turnkey, "or I'll **toss the whole kit and caboodle of you right out**. And first one who lets on to anybody outside how good jail is ain't never getting back in!"

The prisoners instantly fell silent.

The living room of the Schwartz apartment darkened for a moment, as the riot scene faded, and then the face of the announcer appeared, like the sun coming from behind a cloud. "And now, friends," he said, "I have a special message from the makers of anti-gerasone, a message for all you folks over one hundred and fifty. Are you hampered socially by wrinkles, by stiffness of joints and discoloration or loss of hair, all because these things came upon you before anti-gerasone was developed? Well, if you are, you need no longer feel different and out of things."

Busted; v., informal;
broke.
Stilled; v. made quiet.
Gingerly; adv. carefully.

Then, as was the custom for whomever had been most recently disinherited, they began preparing Gramps' breakfast, which would presently be served to him in bed, on a tray. They tried to be cheerful about it. The toughest part of the job was having to handle the honest-to-God eggs and bacon and oleomargarine on which Gramps spent almost all of the income from his fortune.

"Well," said Emerald, "I'm not going to get all panicky until I'm sure there's something to be panicky about."

"Maybe he doesn't know what it was I **busted**," said Lou hopefully.

"Probably thinks it was your watch crystal," said Eddie their son, who was toying apathetically with his buckwheat-type processed sawdust cakes.

"Don't get sarcastic with your father," said Em, "and don't talk with your mouth full, either."

"I'd like to see anybody take a mouthful of *this* stuff and not say something," said Eddie, who was seventy-three. He glanced at the clock. "It's time to take Gramps his breakfast, you know."

"Yeah, it is, isn't it," said Lou weakly. He shrugged. "Let's have the tray, Em."

"We'll both go."

Walking slowly, smiling bravely, they found a large semicircle of long-faced Schwartzes standing around the bedroom door.

Em knocked. "Gramps," she said brightly, "breakfast is ready."

There was no reply, and she knocked again, louder.

The door swung open before her fist. In the middle of the room, the soft, deep, wide, canopied bed, the symbol of the sweet by-and-by to every Schwartz, was empty.

A sense of death, as unfamiliar to the Schwartzes as Zoroastrianism or the causes of the Sepoy Mutiny, **stilled** every voice and slowed every heart. Awed, the heirs began to search **gingerly** under the furniture and behind the drapes for all that was mortal of Gramps, father of the race.

But Gramps had left not his earthly husk but a note, which Lou finally found on the dresser, under a paperweight, which was a treasured souvenir from the 2000 World's Fair. Unsteadily, Lou read it aloud:

" 'Somebody who I have sheltered and protected and taught the best I know how all these years last night turned on me like a mad dog and diluted my anti-gerasone, or tried to. I am no longer a young man. I can no longer bear the crushing burden of life as I once could. So, after last night's bitter experience, I say goodbye. The cares of this world will soon drop away like a cloak of thorns, and I shall know peace. By the time you find this, I will be gone.' "

"Gosh," said Willy brokenly, "he didn't even get to see how the Five-Hundred-Mile Speedway Race was going to come out."

Writing Skills: Summarizing a Short Story



Writing Conventions:

Avoiding Gender Bias and Vague Reference

Gender Bias

In modern English writing, the use of *his* to refer to a group which includes women too is losing its acceptability. This usage is referred to as gender bias. The construction *his* or *her* can be used as a replacement but can be awkward in many cases.

Example

Awkward: Did everyone turn in his or her final draft?

Better: Did everyone turn in the final draft?

Did all the students turn in their final drafts?

Vague Reference

Pronouns such as *which*, *you* and *it*, etc. should have a clear reference in a written text.

Unclear: I forgot what you told me to do, which was unforgivable.

Better: It is unforgivable that I forgot what you told me to do.

Unclear: If you have trouble with spelling, you can benefit from reading more.

Better: People who have trouble with spelling can benefit from reading more.

Unclear: My friends often eat at fast-food restaurants even though they do not really like it

Better: My friends often eat at fast-food restaurants even though they do not really like them.



Focusing on Process/Product Writing

Section 2: Pre-writing Activities

A. Choosing a Topic

Choose a topic for the chapter writing assignment from the following list. Focus on the main organizational types which are indicated below. You will develop your topic into an essay later.

1. Comparison/Contrast/Description: Compare and/or contrast our world today with the world of the characters in "Tomorrow and Tomorrow and Tomorrow."
2. Reason/Result/Description: Kurt Vonnegut, Jr. describes what living in an overpopulated world might be like in the distant future (2158 A.D.). Do you agree or disagree with his vision? Why or why not?
3. Student Generated Topics: If you and your teacher agree on additional topics related to the future or the overpopulation problem, list them and choose the one that interests you most.

B. Brainstorming



C. Paraphrasing/ Summarizing/ Quoting



D. Reporting



E. First Draft



Expanding Your Point of View

Section 1: Peer Feedback

Read your essay to a peer. Follow the "Peer Feedback Guidelines" in your Workbook.

Section 2: Revising

Along with your peer feedback comments, use the following checklist to critically examine your own essay. Your teacher will also use it to evaluate your final draft.

Essay Assessment Guidelines

Content and Ideas

- _____ There is a clear thesis statement.
- _____ There is enough logical support.
- _____ There is unity.

Organization and Form

- _____ There is a clear beginning, middle and end.
- _____ It is clear which are the main supporting points and which are secondary points.

Language Conventions

- _____ There are no fragments, comma splices, run-on sentences.
- _____ There are no problems with subject-verb agreement.
- _____ There is consistency in tense, person, number, and tone.
- _____ There is a variety of sentence types.
- _____ There are no problems with pronoun agreement.
- _____ There is coherence.
- _____ There is clear reference.
- _____ There are no problems with parallelism.
- _____ There is acceptable word choice.
- _____ There are acceptable mechanics (spelling, capitalization, punctuation).

Section 3: Conferencing with the Teacher

Section 4: Further Drafting



Focusing on Process/Product Writing

Section 1: Sharpening Your Editing Skills

When you are satisfied that your final draft is ready for its intended audience, you should reread it several times if possible, looking for errors in grammar, spelling, and punctuation. Two strategies suggested below should help you improve your editing (proofreading) skills.

Strategy 1

1. Read through your text to find sentences that "sound wrong."
2. Read your text again, focusing on the type of errors that you made in your previous final drafts.
Try to focus on one particular type of error that you made each time you reread your draft.

Strategy 2

1. Read through your text to find sentences that "sound wrong."
2. Read your last sentence first and work your way up looking for errors.
3. Reread your draft a third time beginning with your first sentence in order to check your tense sequence.

Strategy 3

When editing someone else's writing, you can use this strategy since you do not know what kind of errors to expect.

1. Read through the text to find sentences that "sound wrong."
2. Look for fragments (incomplete sentences). Make them complete by changing the punctuation or adding more information.
3. Look at how commas are used. Do any of the commas connect sentences together (comma splice)? Correct them.
4. Find all the subjects and verbs. Do they agree with each other in number?
5. Check to see that the verbs are in correct tense and form.



"After years of research, medical science has now developed super anti-gerasone! In weeks, yes weeks, you can look, feel, and act as young as your great-great-grandchildren! Wouldn't you pay \$5,000 to be indistinguishable from everybody else? Well, you don't have to. Safe, tested super-anti-gerasone costs you only dollars a day. The average cost of regaining all the sparkle and attractiveness of youth is less than fifty dollars.

"Write now for your free trial carton. Just put your name and address on a dollar postcard, and mail it 'Super,' Box 500,000, Schenectady, N.Y. Have you got that? I'll repeat it. 'Super' Box..." Underlining the announcer's words was the scratching of Gramps's fountain pen, the one Willy had given him the night before. He had come in a few minutes previous from the Idle Hour Tavern, which commanded a view of Building 257 across the square of asphalt known as Alden Village Green. He had called a cleaning woman to come straighten the place up, and had hired the best lawyer in town to get his descendants a **conviction**. Gramps had then moved the daybed before the television screen so that he could watch from a reclining position. It was something he'd dreamed of doing for years.

"Schen-ec-ta-dy," mouthed Gramps. "Got it." His face had changed remarkably. His facial muscles seemed to have relaxed, revealing kindness and **equanimity** under what had been taut, bad-tempered lines. It was almost as though his trial package of *Super*-anti-gerasone had already arrived. When something amused him on television, he smiled easily, rather than barely managing to lengthen the thin line of his mouth a millimeter. Life was good. He could hardly wait to see what was going to happen next.

Conviction: n. finding someone guilty.

Equanimity: n. calmness of mind.

Oral Discussion

1. What do you think of the ending? Did you anticipate it?
2. The author's ending is a happy one. Can you explain why?

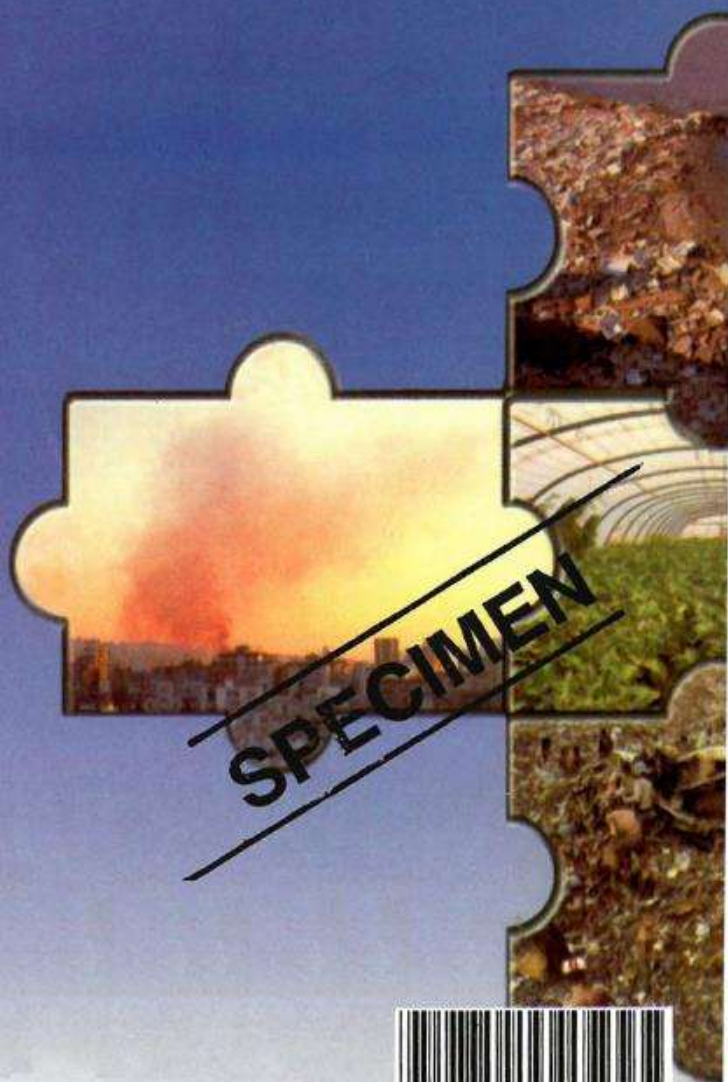
Direction 7: Adding a Reference List

At the end of your paper, you must include a list of all the sources you quoted, paraphrased or summarized. Your teacher will give you the proper format to use.



Direction 8: Presenting an Oral Report

After assessing the final draft of your report, your teacher may ask you for an oral presentation of your report.



SPECIMEN



7SS128

A. Chapter Assessment

Your teacher will assess your essay using the guidelines found in this chapter.

B. Sequenced Research Project**Writing the Final Report****Direction 1: Analyzing Your Material**

Reread all the material that you have written for this project. Since a great deal of time has passed since you first started the project, you may now have a new perspective and want to include material that you previously did not use. Brainstorm this new material using strategies you practiced in your writing assignments.

In order to help yourself analyze all the material you have collected, answer the questions in your Workbook. Be sure to include only the most interesting, informative and convincing material that you have gathered. Always keep your purpose in mind. What are you trying to accomplish by writing this report?

**Direction 2: Organizing Your Information****Direction 3: Writing the Report – First Draft****Direction 4: Checking Your Published Sources****Direction 5: Getting Peer Feedback****Direction 6: Writing the Report – Further Drafting**